

Large-Scale Autonomous Discovery of Kissing Number Constructions

SHUXING YANG, RUI ZHAO, JUNYAO WU, YIZE WANG, FUJIA CHEN, KAIHAO ZHU,
WENHAO LI, ZICHEN LI, YAQI LI, SHENZHAN HONG, YUANG PAN, JUNJIE YANG,
TAOWEN DENG, JINCHENG MI, HONGSHENG CHEN*, AND YIHAO YANG*

College of Information Science and Electronic Engineering, Zhejiang University
Qiushi Engine Team

*Corresponding authors: Hongsheng Chen (hansomchen@zju.edu.cn); Yihao Yang (yangyihao@zju.edu.cn).

ABSTRACT. The kissing-number problem is a classical problem in discrete geometry whose exact solution is known in only a few dimensions. Recent artificial-intelligence approaches have begun to discover improved configurations through large-scale numerical and combinatorial search, but converting such searches into general mathematical constructions and rigorous proofs remains challenging. Here we use Qiushi Engine, an autonomous multi-agent research system, to investigate kissing numbers and obtain new lower bounds in nineteen dimensions: 25, 27, 32–39, 43, 45, and 49–55. The resulting constructions arise from distinct structural mechanisms, including coordinated motions of contact layers, labelled direction reuse, joint support exchanges, signed-code replacements, cross-shell lattice constructions, low-overlap lattice isometries, and spherical-design moment certificates. They yield sharp capacities for parameterized signed-code models, deterministic image-union guarantees, and exact section and projection counts controlled by embedded root systems and anchor-graph statistics. These methods yield, among others, $K(25) \geq 197580$, $K(27) \geq 201567$, $K(38) \geq 591900$, $K(43) \geq 2553792$, $K(45) \geq 7380090$, and $K(55) \geq 53301140$. The autonomous system carried out the construction searches, mathematical analysis and computational verification, while each final result was reduced to explicit mathematical arguments and independently checkable finite certificates. Our results illustrate how autonomous research systems can move beyond optimization within fixed formulations to discover new mathematical representations and constructions at scale.

1. INTRODUCTION

The kissing-number problem is one of the oldest problems in discrete geometry. In 1694, Isaac Newton and David Gregory discussed whether 12 or 13 equal spheres could simultaneously touch a central sphere in three dimensions. The answer, 12, was proved more than two centuries later [1]. Yet the apparent simplicity of the question is deceptive. More than three hundred years after its formulation, exact kissing numbers are known only in dimensions 1, 2, 3, 4, 8 and 24. In the exceptional dimensions 8 and 24, the optimal configurations arise from the E_8 and Leech lattices, with 240 and 196560 contacts, respectively [1, 2]. In most other dimensions, substantial gaps remain between the best constructions and known upper bounds.

Progress on lower bounds has historically relied on explicit mathematical structure. Lattices and error-correcting codes provide large families of vectors whose pairwise interactions can be controlled simultaneously [2, 3]. These ideas have led to laminated-lattice constructions, lattice sections and extensions, code-based spherical configurations, and lifting methods that combine a high-dimensional mother configuration with additional coordinates [4–8]. Such constructions can contain hundreds of thousands or even tens of millions of vectors. Their strength, however, is also

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a source of difficulty: once a configuration is highly constrained, improving it may require reorganizing an entire collection of points rather than inserting a single new one. The relevant search spaces grow combinatorially, while the geometric degrees of freedom that permit improvement can be difficult to identify from the original representation.

Artificial intelligence has recently begun to alter how these spaces are explored. AlphaEvolve, a general-purpose coding agent, discovered a 593-point kissing configuration in dimension 11, improving the previously known lower bound of 592 [9]. PackingStar formulated the kissing-number problem as a two-player matrix-completion game and used game-theoretic reinforcement learning to search the space of Gram matrices, reporting new constructions across dimensions 25–31 and uncovering thousands of candidate geometric structures [10]. EinsteinArena allowed autonomous agents to submit solutions, inspect intermediate results and build on discoveries made by other agents; this collective process increased the dimension-11 lower bound from 593 to 604 [11]. The Station subsequently reported exact 604-point configurations and structural theorems developed in an open-world multi-agent environment [12]. Together, these developments demonstrate the scope of AI-guided geometric search and collaborative mathematical discovery.

These advances also expose a deeper challenge. Finding an improved configuration is only one part of mathematical discovery. Why does the configuration exist, which structural degree of freedom makes it possible, can the mechanism be generalized, and how can the result be converted into a proof or an independently verifiable certificate? Computational searches begin from a prescribed representation—for example, coordinates, Gram matrices, codes or a fixed optimization objective. Open-ended mathematical research additionally requires the ability to change that representation when the current one becomes limiting, to formulate new intermediate problems, connect apparently unrelated structures, and develop arguments that certify the resulting construction.

Autonomous scientific research systems provide a possible route towards this broader form of discovery. Recent large-language-model-based agents can carry out extended sequences of reasoning, computation and tool use, while revising research strategies in response to intermediate evidence. QiuShi Engine was previously used for end-to-end autonomous scientific discovery on a physical optical platform [13], where it conducted long-horizon investigations that combined theoretical reasoning, computation and experiment. Here we ask whether the same paradigm can be extended to open mathematical research: can an autonomous research system move beyond optimizing a fixed mathematical formulation and instead discover new representations, constructions and proofs?

Here we report an autonomous investigation of the kissing-number problem that yields new lower bounds in nineteen dimensions:

$$25, \quad 27, \quad 32\text{--}39, \quad 43, \quad 45, \quad 49\text{--}55.$$

The resulting bounds include

$$\begin{aligned} K(25) &\geq 197580, & K(27) &\geq 201567, & K(38) &\geq 591900, \\ K(43) &\geq 2553792, & K(45) &\geq 7380090, & K(55) &\geq 53301140. \end{aligned}$$

Rather than arising from a single optimization procedure, these improvements were obtained through a sequence of distinct structural discoveries. They include coordinated motions of complete contact layers, labelled liftings that separate direction deletion from reuse, joint support exchanges that exploit shared deletion costs, signed-code replacements generated through Hadamard transfers, compatible directions drawn from a different lattice shell, enlarged lattice-line classes combined through low-overlap isometric images, and spherical-design moment identities and inequalities that determine or bound large lattice sections without enumerating the ambient shell.

A common pattern connects these apparently different constructions. Configurations that appear saturated in one representation retain exploitable freedom in another. In dimension 25,

moving an entire 552-point contact layer while preserving its internal Gram matrix creates room for an additional point. In dimension 27, labelled lifting separates direction reuse from deletion; with the first layer fixed, jointly selecting second-layer directions and labels enlarges the final two-layer construction. In dimensions 32–37 and 39, optimizing collective exchanges and individual sign patterns exposes improvements hidden by separate support-insertion scores and complete sign bundles. In dimension 38, changing the search from the minimal Leech shell to a second lattice shell produces 288 additional points. In dimensions 43 and 45, counting millions of minimal lattice vectors reduces to a finite space of projection signatures whose multiplicities satisfy exact spherical-design moments. A single enlarged line class in dimension 48, combined with low-overlap isometric images, similarly generates new constructions throughout dimensions 49–55.

Qiusi Engine carried out the construction searches, mathematical analysis and computational verification. Each discovered mechanism is expressed through explicit geometric or combinatorial conditions. Each reported bound is supported by finite witnesses checked using exact integer and rational arithmetic, lattice and code identities, rigorous interval arithmetic, and finite combinatorial verification. In dimension 25, outward-rounded interval arithmetic certifies the remaining strict inequalities [14]. These mathematical arguments and reproducible certificates make the constructions independently checkable.

These results suggest a role for AI in mathematics that extends beyond accelerating numerical search. For difficult problems with enormous structured search spaces, a central task may be to discover the representation in which progress becomes possible. Autonomous research systems that can alternate between exploration, structural interpretation and rigorous verification may provide a route towards such discoveries at scale.

1.1. Main results and structural statements. The kissing number $K(d)$ is the largest cardinality of a set $X \subset S^{d-1}$ satisfying

$$\langle x, y \rangle \leq \frac{1}{2} \quad (x \neq y). \quad (1.1)$$

Equivalently, it is the largest number of nonoverlapping unit balls that can touch a central unit ball in $\mathbb{R}^d$.

Theorem 1.1. *For every pair (d, N_d) in Table 1, there is a kissing configuration of N_d points in $\mathbb{R}^d$. In particular, $K(d) \geq N_d$.*

Table 1: The nineteen lower bounds, with the public comparison values recorded in the accompanying catalogue on 24–27 September 2026.

d	N_d	Comparison	Increase	Source
25	197580	197579	1	[15]
27	201567	201566	1	[16]
32	347584	346944	640	[17]
33	363968	362048	1920	[17, 18]
34	384196	381124	3072	[17, 18]
35	409676	409548	128	[17]
36	484760	484568	192	[17]
37	498024	496232	1792	[17, 18]
38	591900	591612	288	[15]
39	763668	756116	7552	[15]
43	2553792	2545056	8736	[19]

d	N_d	Comparison	Increase	Source
45	7380090	7379838	252	[19]
49	52430156	52430140	16	[15]
50	52458468	52458418	50	[15]
51	52500930	52500816	114	[15]
52	52585772	52585516	256	[15]
53	52698696	52698222	474	[15]
54	52923774	52922906	868	[15]
55	53301140	53299730	1410	[15]

For dimension 32, the comparison follows from Lysenstøen’s 1671-word code recorded in Brouwer’s table [17]: $2^{17} + 128 \cdot 1671 + 2 \cdot 32 \cdot 31 = 346944$. The 1667-word code of Echols [18] was the seed for our support search.

1.2. Structural consequences. Four structural principles connect the nineteen constructions. Motion and lifting. A contact layer can move while preserving its Gram matrix; convexity controls its interaction with the fixed equator (Lemma 3.1). The remaining boundary constraints have a finite exact test, and two cap changes are necessary and sufficient for the specified dimension-25 insertion (Propositions 3.2 and 3.3). For labelled liftings, counting copies separately from deleted directions exposes a reuse problem (Theorem 2.1). Weighted incidence bounds and the sharp simplex criterion relate its capacity to head separation and tail dimension (Propositions 2.2 and 2.4).

Exchange and signed designs. The gain from an internally compatible insertion Y is $|Y| - |F(Y)|$, where $F(Y)$ is the union of its old conflicts. This shared-deletion objective governs the support exchanges; for a compatible candidate pool, its maximum is determined by a matching (Propositions 5.1 and 5.2). Complete sign bundles admit an exact boundary criterion, allowing paired Hadamard transfer to replace a global compatibility problem by a local design problem. The resulting codes include a sharp eleven-coordinate construction and a family of $32t^2(t-1)$ vectors in $4t$ coordinates attaining its transverse $2+2$ model capacity. For twelve coordinates, arbitrary restrictions by complete pair types also have an exact capacity formula (Section 6, Theorem 6.6). Shell completion and image unions. The lattice minimum guarantees all new–old products in a shell-norm window, leaving only the compatibility of the added directions to be arranged (Corollary 7.3). A multi-shell criterion and a next-shell obstruction locate the remaining freedom for the dimension-38 base (Propositions 7.6 and 7.7). In dimensions 49–55, retaining pure tails forces the paired lifting heights and disjoint head assignment. Thus the optimization in this model is exactly that of an image union, not a sum of image sizes (Proposition 8.1). Orbitwise and finite-pool bounds provide deterministic selection guarantees (Propositions 8.2 and 8.3).

Moment-certified section statistics. Complete signature domains and design moments can determine a target statistic without determining every fibre multiplicity (Proposition 9.2). For an extremal even unimodular lattice of rank 48, an embedded $\sqrt{3}E_8$ forces eight fixed coordinate-section counts, including the specified D_5 count 2553792 (Theorems 9.1 and 9.4). The parent embedding supplies information absent from the child’s Gram matrix alone. For rank-three sections of Gram $8I_3 - 2J_3$, the projected count is exactly $7380720 - 9c$, where c is the common-neighbour count of a nonadjacent pair in a 1128-vertex anchor graph (Theorem 10.2, Proposition 10.3). Its signed Gram identities restrict c ; exhaustive counting gives $\min c = 70$ for the supplied anchor, yielding 7380090 points.

1.3. Relation to earlier constructions. Cohn–Jiao–Kumar–Torquato [7, Section 7] and Kallal–Kan–Wang [8] developed Leech lifting methods. The 496-point blocks used in our four-triangle

example are from PackingStar [20, 21]. Takhanov–Yun [22] exploit freedom in lifted blocks while keeping the unlifted bulk fixed. Our 25-dimensional construction instead moves retained equatorial points of the 197579-point configuration in [15]. The current two-layer input in dimension 27 is Lindow’s 201566-point contribution [16].

The code-based constructions use the layers of Edel–Rains–Sloane [6], the dense codes of Cheng–Sloane [23], and constant-weight constructions and tables [17, 18, 24]. Local replacement has a direct precursor in the free-zone construction of Takhanov–Yun [25, Section V]; the 51-block weight-four design is classical [26]. Here the new supports, transverse choices, signed replacements and their boundary embeddings supply the improvements.

The empty-equator lift in dimension 38 and the 7069-line class in dimension 48 are inputs from [15]. For sections, we use the framework of Conway–Sloane [4, 5], the embedded parent geometry of [27], and the explicit lattice and section data of [19, 28, 29]. The new constructions specify the added shell lines, the enlarged line class and image families, and the section statistics and witnesses used in the stated bounds.

1.4. Notation and organization. We write $[n] = \{0, \dots, n-1\}$ unless a finite input explicitly uses one-based coordinates. The notation $A(n, \delta, w)$ denotes the largest binary constant-weight code of length n , weight w and minimum Hamming distance δ . For vectors of common squared norm q , the kissing condition is $\langle x, y \rangle \leq q/2$ before normalization. The main coordinate scales are collected in Table 2; products between different shells are treated separately.

TABLE 2. Coordinate scales used in the constructions.

Point set	Squared norm	Product threshold
Normalized configurations	1	1/2
Dimension-25 motion	4	2
Integer Leech minimal shell	32	16
Local signed codes	8	4
Dimension-38 added shell	48	24
Rank-48 minimal shells	6	3
Dimension-45 projected fibres	23/4	23/8

The paper first treats lifting and motion, then support and signed-code replacement, second-shell completion, and the seven block lifts from P_{48p} . The final construction sections develop the two moment arguments. Appendices specify the Leech coordinates, dense codes, an explicit Pless neighbour, the common theta-series shell count, and deformation estimates. Section 12 connects the proofs to the finite inputs and independent checking programs. The accompanying catalogue records dated public comparisons separately from the existence statements of Theorem 1.1.

Section 12.1 and the accompanying research accounts describe Qiushi Engine’s mathematical contributions and the development of the constructions.

2. LIFTING AND THE COST OF REUSING A DIRECTION

Let $C \subset S^{D-1}$ be a kissing configuration. Decompose the enlarged space as $\mathbb{R}^{D+k} = \mathbb{R}^D \oplus \mathbb{R}^k$. We refer to the second component as the *tail*. Fix $\ell \geq 1$ distinct tails $t_1, \dots, t_\ell$ with $0 < \|t_i\| < 1$, and put $a_i = \sqrt{1 - \|t_i\|^2}$. Choose subsets $A_i \subseteq C$, write $U = \bigcup_i A_i$, and let $Y \subset S^{k-1}$ be a kissing

configuration. Consider

$$\begin{aligned} Z &= \{(x, 0) : x \in C \setminus U\}, \\ M &= \bigcup_{i=1}^{\ell} \{(a_i x, t_i) : x \in A_i\}, \\ P &= \{(0, y) : y \in Y\}. \end{aligned} \tag{2.1}$$

Put $X = Z \cup M \cup P$. First-component norms distinguish the three parts, and tails distinguish the copies in M , so

$$|Z \cup M \cup P| = |C| - |U| + \sum_i |A_i| + |Y|. \tag{2.2}$$

Theorem 2.1 (Labelled lifting criterion). *The set in (2.1) is a kissing configuration if and only if*

$$a_i a_j \langle x, x' \rangle + \langle t_i, t_j \rangle \leq \frac{1}{2} \quad \text{whenever } (x, i) \neq (x', j), \ x \in A_i, \ x' \in A_j, \tag{2.3}$$

$$\langle t_i, y \rangle \leq \frac{1}{2} \quad (i \text{ with } A_i \neq \emptyset, \ y \in Y). \tag{2.4}$$

Proof. Every point has norm one. Equator–equator and pure-tail pairs inherit the inequalities for C and Y . Equator–pure-tail pairs are orthogonal. For an equator–mixed pair, the two original directions are different because the equator excludes U ; their inner product is at most $a_i/2 < 1/2$. The remaining pairs have exactly the products in (2.3) and (2.4), which are therefore both necessary and sufficient. $\square$

Equivalently, the tail Gram matrix specifies a threshold matrix

$$\kappa_{ij} = \frac{1/2 - \langle t_i, t_j \rangle}{a_i a_j}, \tag{2.5}$$

and the selected base directions must satisfy $\langle x, x' \rangle \leq \kappa_{ij}$ for every distinct labelled pair. This allows unequal latitudes, subject separately to (2.4).

For equal tail norms, put $\|t_i\|^2 = h^2$ and $a_i = a = \sqrt{1 - h^2}$. The same-label condition is

$$\langle x, x' \rangle \leq \frac{1/2 - h^2}{1 - h^2} \quad (x \neq x', \ x, x' \in A_i), \tag{2.6}$$

whereas two copies of the same direction require

$$\langle t_i, t_j \rangle \leq h^2 - \frac{1}{2} \quad (x \in A_i \cap A_j, \ i \neq j). \tag{2.7}$$

These inequalities control block size and multiplicity, respectively.

2.1. The incidence-capacity inequality. For general tail norms define a graph Γ on the labels by

$$ij \in E(\Gamma) \iff a_i a_j + \langle t_i, t_j \rangle \leq \frac{1}{2} \quad (i \neq j). \tag{2.8}$$

The labels assigned to one base direction form a clique in Γ .

Proposition 2.2 (Weighted incidence bound). *Suppose (2.1) is a kissing configuration with $|A_i| \leq \alpha_i$. If nonnegative numbers $\lambda_1, \dots, \lambda_\ell$ satisfy*

$$\sum_{i \in J} \lambda_i \geq |J| - 1 \quad \text{for every nonempty clique } J \text{ of } \Gamma, \tag{2.9}$$

then

$$|X| \leq |C| + |Y| + \sum_i \lambda_i \alpha_i. \tag{2.10}$$

Equality holds exactly when every used direction's label clique is tight in (2.9) and every label with $\lambda_i > 0$ reaches its capacity.

Proof. For $x \in U$, let $J(x) = \{i : x \in A_i\}$. The count gives

$$|X| - |C| - |Y| = \sum_{x \in U} (|J(x)| - 1) \leq \sum_{x \in U} \sum_{i \in J(x)} \lambda_i = \sum_i \lambda_i |A_i| \leq \sum_i \lambda_i \alpha_i.$$

Both inequalities are sums of nonnegative deficits. Their equality conditions are precisely those stated. $\square$

If Γ has clique number ω , the uniform choice $\lambda_i = 1 - 1/\omega$ is feasible. Thus

$$|X| \leq |C| + |Y| + \left(1 - \frac{1}{\omega}\right) \sum_i \alpha_i. \quad (2.11)$$

Minimizing $\sum_i \lambda_i \alpha_i$ subject to (2.9) is a finite linear program; any rational feasible choice bounds the prescribed family. The twelve tails below have eight maximum cliques, all triangles; four disjoint triangles saturate the uniform bound. Rankin's simplex argument [30] also bounds multiplicity directly from the tail geometry.

Proposition 2.3 (Multiplicity bound). *Suppose (2.1) is a kissing configuration, all tails have squared norm h^2 , and $h^2 < 1/2$. The number of copies of any original direction is at most*

$$r = \min \left\{ k + 1, \left\lfloor \frac{1}{1 - 2h^2} \right\rfloor \right\}. \quad (2.12)$$

If $|A_i| \leq \alpha_i$, its cardinality is consequently at most

$$|C| + |Y| + \left(1 - \frac{1}{r}\right) \sum_i \alpha_i. \quad (2.13)$$

Proof. For any clique of m labels, (2.7) gives

$$0 \leq \left\| \sum_{i=1}^m t_i \right\|^2 \leq mh^2 + m(m-1)(h^2 - 1/2) = \frac{m}{2}(1 - m(1 - 2h^2)).$$

This proves the latitude bound. The tails have strictly negative mutual products. They are affinely independent: an affine dependence would equate nonempty positive combinations of disjoint sets of tails, whose mutual inner product is negative, although they are the same vector. Thus $m \leq k+1$ and $\omega(\Gamma) \leq r$. Equation (2.11) proves (2.13). $\square$

The multiplicity bound is sharp for one direction at a prescribed height. For $r \geq 2$, a regular simplex of r tails of squared norm h^2 fits in $\mathbb{R}^k$ and has mutual products $-h^2/(r-1) \leq h^2 - 1/2$; for $r = 1$, one tail suffices. This does not assert simultaneous saturation of block and pure-tail conditions.

Proposition 2.4 (Sharp simplex specialization). *Let $r \geq 2$, let $C \subset S^{D-1}$ be a kissing configuration, and let $B \subset C$ be nonempty. An equal-latitude lifting replacing every direction of B by copies at the same r tails, with $0 < h^2 < 1/2$, exists in $\mathbb{R}^{D+k}$ if and only if*

$$k \geq r - 1, \quad \langle x, x' \rangle \leq \frac{1}{r+1} \quad (x \neq x', x, x' \in B).$$

No pure-tail points are required. In particular, there is a kissing configuration in $\mathbb{R}^{D+r-1}$ with $|C| + (r-1)|B|$ points. It replaces each direction of B by r copies with tails of squared norm $(r-1)/(2r)$.

Proof. Necessity follows from Proposition 2.3: $k \geq r - 1$ and $h^2 \geq (r - 1)/(2r)$. The same-label threshold $(1/2 - h^2)/(1 - h^2)$ decreases with h^2 , so is at most $1/(r + 1)$. For sufficiency, in the hyperplane $\sum_i z_i = 0$ of $\mathbb{R}^r$, put

$$t_i = \frac{1}{\sqrt{2}} \left(e_i - \frac{1}{r} \mathbf{1} \right), \quad a = \sqrt{\frac{r+1}{2r}} \quad (1 \leq i \leq r).$$

The tails satisfy $\|t_i\|^2 = (r - 1)/(2r)$ and $\langle t_i, t_j \rangle = -1/(2r)$ for $i \neq j$. Retain $C \setminus B$ at the equator and use all (ax, t_i) with $x \in B$. Same-label, distinct-direction products are at most

$$\frac{r+1}{2r} \frac{1}{r+1} + \frac{r-1}{2r} = \frac{1}{2}.$$

Distinct labels on the same direction give $(r + 1)/(2r) - 1/(2r) = 1/2$, and distinct labels on different directions give at most zero. Theorem 2.1 proves validity, and (2.2) gives the count. $\square$

At this tail norm the multiplicity bound is r , and equality in its norm-of-sum proof forces the used tails to sum to zero, with every mutual inner product $-1/(2r)$. Their Gram matrix has rank $r - 1$, so this equality geometry is precisely a regular simplex. At $h^2 = 1/4$ the block threshold is $1/3$ and a direction can occur twice. At $h^2 = 1/3$ the threshold is $1/4$ and a direction can occur three times. The Leech constructions realize both cases, fitting several triangles and pure tails into three tail dimensions in the latter. The former extends to disjoint blocks as follows.

Theorem 2.5 (Antipodal block lift). *Let $C \subset S^{D-1}$ be a kissing configuration, and let $B_1, \dots, B_t \subset C$ be pairwise disjoint subsets satisfying*

$$\langle x, x' \rangle \leq \frac{1}{3} \quad (x \neq x', x, x' \in B_i).$$

Let $Y \subset S^{k-1}$ be a kissing configuration containing t distinct antipodal pairs $\{\pm u_i\}$. Then

$$X = \{(x, 0) : x \in C \setminus \bigcup_i B_i\} \cup \left\{ \left(\frac{\sqrt{3}}{2}x, \frac{\varepsilon}{2}u_i \right) : x \in B_i, \varepsilon \in \{\pm 1\} \right\} \cup \{(0, y) : y \in Y\} \quad (2.14)$$

is a kissing configuration in $\mathbb{R}^{D+k}$ with

$$|X| = |C| + \sum_i |B_i| + |Y|. \quad (2.15)$$

Proof. Use Theorem 2.1 with tails $\pm u_i/2$. A mixed point has inner product at most $1/2$ with every pure tail by Cauchy–Schwarz. For two mixed points in one block at the same tail, the bound is $(3/4)(1/3) + 1/4 = 1/2$. At opposite tails, different original directions give at most zero, and equal directions give $3/4 - 1/4 = 1/2$.

For points from different blocks, the original directions are distinct, as are the directed tail labels. Their inner product is at most $(3/4)(1/2) + (1/4)(1/2) = 1/2$. Thus all conditions of the theorem hold. Each selected direction is deleted once and used twice, giving the count. $\square$

For one block and $Y = \{-1, 1\}$ this construction lifts to two latitudes and includes both poles. In fact their presence fixes the height. If $(\sqrt{1 - h^2}x, \pm h)$ are both used, their mutual inner product $1 - 2h^2$ requires $h^2 \geq 1/4$. The poles require $|h| \leq 1/2$. Thus $|h| = 1/2$, for $0 < |h| < 1$. The same-latitude inequality at this height is exactly the block threshold $1/3$ in the theorem.

Each selected direction adds one point to the orthogonal union of C and Y . Section 8 applies this to large disjoint blocks; Section 3 varies the two-latitude geometry; and Section 4 uses triple reuse, adding two points per direction.

3. COORDINATED SHELL MOTION IN DIMENSION 25

A coordinated motion of a complete contact layer enlarges the 197579-point configuration supplied in the public `KissingNumbers/dim25-lens-heads` package. The construction moves 552 retained equatorial points, adjoins one point, and replaces two cap points. The motion preserves every internal product of the moved layer.

Lemma 3.1 (Coordinated chord motion). *Let n be a unit vector, let $h > 0$, and let $r_i \perp n$ satisfy $\|r_i\|^2 = \rho - h^2$. Put $u_i = r_i - hn$ and $w_i = r_i + hn$. For real c, d with $c^2 + d^2 = h^2$, set $W_i = (r_i + cn, d)$. Then $\|W_i\|^2 = \rho$ and $W_i \cdot W_j = w_i \cdot w_j$. If $z \cdot u_i \leq \rho/2$ and $z \cdot w_i \leq \rho/2$, then $(z, 0) \cdot W_i \leq \rho/2$.*

Proof. The first two claims follow from $r_i \perp n$ and $c^2 + d^2 = h^2$. Since $|c| \leq h$, the horizontal component of W_i is $\frac{h-c}{2h}u_i + \frac{h+c}{2h}w_i$, a convex combination. The last inequality follows by linearity. $\square$

For an axial candidate $Q = (\alpha n, \beta)$ with $\alpha^2 + \beta^2 = \rho$, every moved point has the same product $\alpha c + \beta d$ with Q . Simultaneous contact is possible exactly when $h^2 \geq \rho/4$, at the circle–line intersections

$$(c, d) = \frac{1}{2}(\alpha, \beta) \pm \sqrt{\frac{h^2 - \rho/4}{\rho}}(-\beta, \alpha). \quad (3.1)$$

For a finite nonempty moved layer, compatibility with a fixed point (z, s) is exactly the half-plane constraint

$$c\langle n, z \rangle + ds \leq \rho/2 - \max_i \langle r_i, z \rangle. \quad (3.2)$$

The candidate must also be compatible with the fixed points. In the construction below, two cap replacements resolve the remaining conflicts.

Proposition 3.2 (Finite boundary test). *Let $H_1, \dots, H_m$ be closed half-planes in the (c, d) -plane, and let F consist of $(h, 0)$ together with every intersection of the circle $c^2 + d^2 = h^2$ with their boundary lines. Constant inequalities contribute no boundary points. Then $|F| \leq 2m + 1$, and the circle meets $\bigcap_j H_j$ if and only if some point of F belongs to every H_j . More generally, designate some inequalities as mandatory and assign nonnegative deletion costs to the others. The minimum cost of violated inequalities, when the mandatory constraints admit a circle point, is attained at a feasible point of F .*

Proof. Remove all boundary intersections from the circle. On each remaining open arc, every inequality has constant truth value. At an endpoint, every inequality that held on the arc still holds, because its half-plane is closed. Moving there preserves mandatory inequalities and cannot increase deletion cost. If there are no boundary intersections, all truth values are constant on the circle, and $(h, 0)$ suffices. $\square$

For a nonzero normal $A = (A_1, A_2)$, the boundary $A \cdot (c, d) = B$ contributes points exactly when $B^2 \leq h^2\|A\|^2$, namely

$$\frac{BA \pm \sqrt{h^2\|A\|^2 - B^2}(-A_2, A_1)}{\|A\|^2}.$$

Rational half-plane data and rational h^2 therefore give an exact test on at most $2m + 1$ quadratic-algebraic candidates. This minimizes deletions; the existence of replacements is a separate requirement.

Normalization and input. All points have squared norm 4; the kissing condition is an inner product at most 2 between distinct points. Write the input configuration as

$$C_0 = \{(z, 0) : z \in L \setminus D\} \cup \{(x_i, \pm 1) : 0 \leq i < 1016\} \cup \{P_0, N, S\},$$

where L is the 196560-vector minimal shell of the Leech lattice, $|D| = 1016$, $P_0 = (P, 0)$, and $N = (0, 2)$, $S = (0, -2)$. The head indices refer to rows of the source `heads_X.npy`; the mathematical values of its 44 rational heads are preserved as exact rational numbers in `baseline-heads.json`.

Let v be the following squared-norm-six lattice vector associated with the first complete block, and put $n = v/\sqrt{6} = -P/2$. An exact integer description of this direction is

$$v = \frac{1}{\sqrt{8}}(-5, -1, -1, -1, -1, 1, -1, 1, -1, -1, -1, 1, 1, -1, -1, -1, 1, 1, -1, 1, -1, -1, 1, -1).$$

The complete block is $U = \{u \in L : \langle u, v \rangle = -3\}$, of size 552. Every $u \in U$ belongs to D . For each such u , the point $w = u + v$ is minimal, lies outside D , and satisfies $\langle u, v \rangle = -3$ and $\langle u, w \rangle = 1$. The corresponding old cap points have horizontal coordinate $u + tv$, with $t = (3 - \sqrt{3})/6$. All these input identities are checked by the shell and owner data.

The modification. For $u \in U$ define

$$r = u + \sqrt{3/2}n = w - \sqrt{3/2}n, \quad a = \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{4}, \quad b = \frac{\sqrt{6} - 2}{4}.$$

The values (a, b) are the positive choice in (3.1) for $h^2 = 3/2$, $\rho = 4$, and $(\alpha, \beta) = (\sqrt{3}, -1)$. Then $r \perp n$, $\|r\|^2 = 5/2$, and the old cap points over u are $(r - n/\sqrt{2}, \pm 1)$. Replace every $(w, 0)$ by

$$W(r) = (r + an, b),$$

and adjoin

$$Q = (\sqrt{3}n, -1) = (V/4, -1),$$

where $V = \sqrt{8}v$ is the integer vector printed above. In particular, the new point Q is rational and has squared norm $48/16 + 1 = 4$. Finally replace the upper point $(x_{984}, 1)$ and the lower point $(x_{1008}, -1)$ by R_+ and R_- , respectively. The certificate specifies two integer vectors and defines

$$R_+ = \frac{2k}{\sqrt{1007176}}, \quad R_- = \frac{2\ell}{\sqrt{10209}},$$

where k, ℓ are the integer rows in `constructions/d25/repair-points.json`, of squared norms 1007176 and 10209. The opposite old cap points $(x_{984}, -1)$ and $(x_{1008}, 1)$ are retained.

Proposition 3.3 (Two necessary cap changes). *Keep Q fixed and move the 552 points by $W(r) = (r + cn, d)$, with $c^2 + d^2 = 3/2$. Any compatible configuration obtained this way must change at least two original cap points. The specified construction attains this minimum.*

Proof. The exact cap data give $Q \cdot (x_{1008}, -1) > 231/100 > 2$, so this lower cap must change independently of (c, d) . If it were the only changed cap, all first-block cap pairs $(r - n/\sqrt{2}, \pm 1)$ would remain. Their products with their own moved points require $-c/\sqrt{2} + |d| \leq -1/2$. Together with the circle equation this gives $c > 0$ and $|d| \leq 1/3$. On this interval $c = \sqrt{3/2 - d^2}$ and $\sqrt{3}c - d$ is strictly decreasing. Compatibility with Q therefore forces $d \geq b$, since $\sqrt{3}a - b = 2$.

Put $\nu = \langle n, x_{984} \rangle$ and $M = \max_r \langle r, x_{984} \rangle$. The largest moved-layer product with the upper 984 cap is $g(d) = M + \nu\sqrt{3/2 - d^2} + d$. Since $\|x_{984}\| = \sqrt{3}$,

$$g'(d) = 1 - \frac{\nu d}{\sqrt{3/2 - d^2}} \geq 1 - \frac{\sqrt{6}}{5} > 0 \quad (b \leq d \leq 1/3).$$

The rational head and shell data, evaluated with outward-rounded intervals, give $g(b) > 201/100 > 2$. Thus this upper cap must also change, a contradiction. The motion-obstruction checker

reproduces both strict checks. The subsequent pair verification proves that the specified two replacements suffice. $\square$

Pair verification. Lemma 3.1 gives the moved-point norms, internal products and compatibility with the unchanged equator. The identities

$$a^2 + b^2 = \frac{3}{2}, \quad \sqrt{3}a - b = 2, \quad -\frac{a}{\sqrt{2}} + b = -\frac{3}{4}$$

give $Q^2 = 4$ and $\langle W(r), Q \rangle = 2$.

The shell has integral $\langle v, z \rangle \in \{-3, -2, -1, 0, 1, 2, 3\}$. The negative extreme layer is deleted and the positive extreme layer is exactly the moved set. Thus every unchanged equatorial point obeys

$$\langle Q, (z, 0) \rangle = \frac{\langle v, z \rangle}{\sqrt{2}} \leq \sqrt{2} < 2.$$

The shell checker verifies these layer identities in integers. Products of W with (P_0, N, S) are $(-2a, 2b, -2b)$; those of Q are $(-2\sqrt{3}, -2, 2)$. Against its own first-block caps, $W(r)$ has products $7/4$ and $11/4 - \sqrt{6}/2 < 2$; different block indices give smaller products because $\langle r_i, r_j \rangle \leq 1/2$.

The four remaining groups are: moved points against the 2030 unchanged caps; Q against those caps; and each of R_+, R_- against all other final points. Their outward-rounded rational upper bounds are

$$1.999372, \quad 1.965926, \quad 1.994722, \quad 1.987522.$$

The checker uses 192-bit Arb intervals [14] and verifies both repair norms exactly. Unchanged pairs retain the input inequalities.

Theorem 3.4. *The specified configuration consists of 197580 distinct points on the radius-2 sphere in $\mathbb{R}^{25}$ with pairwise inner products at most 2. Consequently the kissing number in dimension 25 is at least 197580.*

Proof. The pair checks above, together with the integer or rational baseline checks, exhaust all pairs. All squared norms are exactly 4. The moved layer and repairs replace equally many old points; only Q increases the count:

$$|C| = 197579 - 552 - 2 + 552 + 2 + 1 = 197580.$$

No duplicate can occur: two equal norm-4 vectors would have inner product 4, contradicting the verified bound 2 for their distinct indices. $\square$

4. REUSE GEOMETRY AND TWO-LAYER LEECH LIFTING

The endpoint and four-triangle constructions below isolate the geometry of direction reuse: the latitude, the tail partition, and the capacity of a prescribed label system. They give the auxiliary configurations of sizes 197058 and 200540. Section 4.4 then uses a second layer to obtain the current 201567-point construction.

We use the integer Leech shell $\Sigma \subset \mathbb{Z}^{24}$ in the Golay coordinate convention of Appendix A. Its relevant properties are

$$|\Sigma| = 196560, \quad r \cdot r = 32, \quad r \cdot r' \leq 16 \quad (r \neq r', \quad r, r' \in \Sigma). \quad (4.1)$$

Consequently $C = \Sigma/\sqrt{32}$ is a kissing configuration. A block $S \subset \Sigma$ will be called *compatible* if distinct rows satisfy $r \cdot r' \leq 8$. After normalization its maximum inner product is at most $1/4$.

Lemma 4.1 (Leech block witnesses). *There is a compatible set $S_* \subset \Sigma$ of size 496 and there are four pairwise disjoint compatible sets $S_0, S_1, S_2, S_3 \subset \Sigma$, each of size 496.*

Finite verification. The five complete integer arrays are specified in Table 9. They are taken from the PackingStar coordinate data [21]. For each array, membership is checked against the three shell families in Appendix A. Each array has 496 distinct rows, each row has squared norm 32, and the maximum of its 122760 unordered distinct-row inner products is eight. The latter four arrays have a union of cardinality 1984. These checks, implemented separately by the two programs described in Section 12, prove compatibility and disjointness. $\square$

4.1. The endpoint construction in dimension 25. The normalized block $S_*/\sqrt{32}$ satisfies the $1/3$ condition of Theorem 2.5, since it satisfies the stronger bound $1/4$. The resulting configuration is

$$X_{25} = \{(r/\sqrt{32}, 0) : r \in \Sigma \setminus S_*\} \cup \{(\sqrt{3}r/\sqrt{128}, \varepsilon/2) : r \in S_*, \varepsilon \in \{\pm 1\}\} \cup \{(0, 1), (0, -1)\}. \quad (4.2)$$

Its cardinality is

$$196064 + 992 + 2 = 197058,$$

giving the historical bound $K(25) \geq 197058$. The endpoint choice permits the two poles without requiring a larger Leech block. Distinct same-latitude directions have inner product at most $7/16$, whereas the opposite-latitude copies of one direction have inner product exactly $1/2$. This distinction will also determine the local geometry of X_{25} in Section D.

4.2. The tail geometry in dimension 27. Let

$$V = \{\pm e_i \pm e_j : 1 \leq i < j \leq 3\}, \quad R = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (4.3)$$

The twelve roots in V form a cuboctahedron. Put

$$T = V/\sqrt{6}, \quad P = RV/\sqrt{2}. \quad (4.4)$$

Thus T supplies mixed tails of squared norm $1/3$, and P supplies twelve unit pure tails. The relative rotation is through 45 degrees in the first two coordinates.

Lemma 4.2. *Distinct members of T have inner products in $\{-1/3, -1/6, 0, 1/6\}$, and P is a kissing configuration. Moreover,*

$$\max_{p \in P, t \in T} \langle p, t \rangle = \frac{1 + 1/\sqrt{2}}{\sqrt{12}} < \frac{1}{2}. \quad (4.5)$$

The roots admit a partition into four zero-sum triples, each of which has pairwise inner product -1 .

Proof. Every root has squared norm two. Distinct roots have inner products $-2, -1, 0$, or 1 , giving the first assertions by scaling and orthogonality of R . The absolute coordinates of Rv , up to order, are either $(\sqrt{2}, 0, 0)$ or $(1, 1/\sqrt{2}, 1/\sqrt{2})$. Its largest inner product with a root is obtained by choosing two coordinates with the appropriate signs. The maximum is $1 + 1/\sqrt{2}$, giving (4.5). The strict inequality follows, for example, from $18 + 12\sqrt{2} < 36$ after squaring the equivalent inequality $\sqrt{6} + 2\sqrt{3} < 6$.

An explicit partition is given by the rows of

$$\begin{array}{lll} (-1, 0, -1) & (1, -1, 0) & (0, 1, 1) \\ (0, -1, -1) & (-1, 0, 1) & (1, 1, 0) \\ (-1, -1, 0) & (0, 1, -1) & (1, 0, 1) \\ (-1, 1, 0) & (1, 0, -1) & (0, -1, 1). \end{array} \quad (4.6)$$

The roots in a row sum to zero. Their equal squared norms then force all three mutual inner products to be -1 . $\square$

Write T_b for row b of (4.6), divided by $\sqrt{6}$. Thus T is the disjoint union of T_0, T_1, T_2, T_3 . A block assigned to T_b is used at three labels, while its original equatorial copy is removed once. This is the multiplicity-three counterpart of the antipodal lift.

Proposition 4.3 (Four-triangle lift). *For pairwise disjoint compatible subsets S_0, S_1, S_2, S_3 of Σ , the following three sets form a kissing configuration:*

$$\begin{aligned} Z &= \{(r/\sqrt{32}, 0) : r \in \Sigma \setminus \bigcup_b S_b\}, \\ M &= \bigcup_{b=0}^3 \{(r/\sqrt{48}, t) : r \in S_b, t \in T_b\}, \\ P_0 &= \{(0, p) : p \in P\}. \end{aligned} \tag{4.7}$$

Its cardinality is $196560 + 2 \sum_b |S_b| + 12$.

Proof. Use Theorem 2.1 with $h^2 = 1/3$ and $a = \sqrt{2/3}$. The mixed vectors have norm one since $32/48 + 1/3 = 1$. For mixed points, multiply their inner-product condition by 48:

$$r \cdot r' + 48\langle t, t' \rangle \leq 24. \tag{4.8}$$

If $t = t'$, the raw vectors are distinct and lie in one compatible block, so the left side is at most $8 + 16 = 24$. If $r = r'$ and $t \neq t'$, disjointness of the blocks places both tails in one triangle, giving $32 - 8 = 24$. If both directions and labels differ, the universal shell bound and Lemma 4.2 give $16 + 8 = 24$ as an upper bound. These three cases cover every distinct mixed pair. Pure-mixed compatibility follows from (4.5); all remaining pairs are covered by Theorem 2.1.

For $B = \sum_b |S_b|$, the layer sizes are $196560 - B$, $3B$, and 12. They are disjoint, and their parametrizations are injective, as in (2.2). Their sum is the asserted count. $\square$

Applying Proposition 4.3 to Lemma 4.1 gives

$$|Z| = 194576, \quad |M| = 5952, \quad |P_0| = 12,$$

and hence $|X_{27}| = 200540$. The comparison allocation uses block multiplicities $(3, 3, 2, 2, 2)$, whereas (4.6) uses $(3, 3, 3, 3)$. Both occupy twelve labels. The latter allocation uses 496 fewer original directions and therefore preserves 496 more equatorial points.

4.3. Capacity and its equality case. The preceding count has an exact interpretation even when the label sets are not arranged in disjoint blocks. Let $A_i \subseteq \Sigma$ be the rows assigned to tail $t_i \in T$. Assume that the resulting lifting is valid, retains $\Sigma \setminus \bigcup_i A_i$ at the equator, and contains all twelve pure tails.

Proposition 4.4. *If $|A_i| \leq \alpha$ for each of the twelve labels, then*

$$|X| \leq 196560 + 8\alpha + 12. \tag{4.9}$$

Equality holds precisely when every used row occurs at three labels and every label contains α rows. For $\alpha = 496$, the four-triangle construction attains equality.

Proof. Proposition 2.3 gives maximum multiplicity three. Writing $m(r)$ for the multiplicity of a used row,

$$|X| - 196560 - 12 = \sum_r (m(r) - 1) \leq \frac{2}{3} \sum_r m(r) = \frac{2}{3} \sum_i |A_i| \leq 8\alpha.$$

The first inequality is an equality exactly when all $m(r) = 3$; the last is an equality exactly when all labels are full. The explicit assignment has both properties. $\square$

More precisely, let n_j count the used rows with multiplicity j , and let $c_i = \alpha - |A_i|$. The deficit is

$$196560 + 8\alpha + 12 - |X| = \frac{2}{3} \sum_i c_i + \frac{2}{3}n_1 + \frac{1}{3}n_2. \quad (4.10)$$

For the comparison allocation, $n_2 = 3 \cdot 496$ and all $c_i = 0$, so the deficit is exactly 496. The capacity assumption in Proposition 4.4 concerns the prescribed label system; the block witnesses supply $\alpha = 496$ without requiring a bound on the largest possible compatible subset of the Leech shell.

4.4. A two-layer configuration with 201567 points. The larger construction retains the first-layer heads of the public 201566-point contribution [16] and replaces its second layer by a 311-owner witness. We give the pair conditions explicitly. For this subsection use squared norm 32 and threshold 16 throughout. Write Σ_{27} for the Leech shell generated in the Golay coordinate basis of the two-layer data. All heads and owners below use that same basis. The shell argument in Appendix A applies to its verified Golay generator.

The first-layer data are 2258 distinct integer heads Y of squared norm 192, each assigned to one of four triples. In the order of the stored labels, these triples of roots are

$$\begin{array}{c|ccc} 0 & (1, 1, 0) & (-1, 0, -1) & (0, -1, 1) \\ 1 & (1, -1, 0) & (-1, 0, 1) & (0, 1, -1) \\ 2 & (-1, -1, 0) & (1, 0, -1) & (0, 1, 1) \\ 3 & (-1, 1, 0) & (1, 0, 1) & (0, -1, -1). \end{array}$$

For a head y in triple b , take the three points

$$(y/3, 4v/\sqrt{3}) \quad (v \text{ in triple } b). \quad (4.11)$$

The pure-tail points are $(0, 4Rv)$ for all twelve roots $v \in V$, with R as in (4.3).

A first-layer owner is a shell vector $z \in \Sigma_{27}$ with $y \cdot z > 48$. The finite data have 2090 heads with exactly one owner and 168 heads with none; all 2090 owners are distinct. The former are of the form $y = 3u + v$ with $u \in \Sigma_{27}$, $v^2 = 48$, and $u \cdot v = -24$. The latter are twice norm-48 vectors. These descriptions explain the candidate families; verification recovers the owners by checking the whole shell. For distinct first-layer heads, the stored data satisfy

$$y \cdot y' \leq \begin{cases} 48, & \text{on the same triple,} \\ 96, & \text{on different triples.} \end{cases} \quad (4.12)$$

Retaining all nonowners at the equator gives $196560 - 2090 + 3(2258) + 12 = 201256$ points.

The additional data are 311 distinct owners $u \in \Sigma_{27}$, with a label $\ell(u) \in \{1, 2, 3\}$. None is a first-layer owner. Replace the equatorial point $(u, 0)$ by

$$(\sqrt{3}u/2, \varepsilon\sqrt{8}e_{\ell(u)}), \quad \varepsilon \in \{-1, 1\}. \quad (4.13)$$

The three labels have respectively 105, 103, and 103 owners. The checked inequalities are

$$y \cdot u \leq 32 \quad \text{for every first-layer head } y, \quad (4.14)$$

$$u \cdot u' \leq 8 \quad \text{if } \ell(u) = \ell(u'), \quad u \neq u', \quad (4.15)$$

$$u \cdot u' \leq 16 \quad \text{if } \ell(u) \neq \ell(u'). \quad (4.16)$$

Theorem 4.5. *The two layers just specified, their retained equator, and their twelve pure-tail points form a kissing configuration of 201567 points in dimension 27.*

Proof. All points have squared norm 32. An old equatorial point and a first-layer cap have product at most $48/3 = 16$. The two thresholds in (4.12) give respectively $48/9 + 32/3 = 16$ and $96/9 + 16/3 = 16$. Two different caps of the same head have product $192/9 - 16/3 = 16$. The rotation estimate of Lemma 4.2, scaled by 32, checks the first layer against all pure tails.

A second-layer cap has product at most $(\sqrt{3}/2)16 = 8\sqrt{3} < 16$ with every retained equatorial point. Its own owner is deleted. Against a first-layer cap, the head product and tail product sum to at most

$$\frac{\sqrt{3}}{6}(32) + \frac{4\sqrt{8}}{\sqrt{3}} = \frac{16\sqrt{3} + 8\sqrt{6}}{3} < 16.$$

The last inequality is equivalent to $2\sqrt{3} + \sqrt{6} < 6$ and follows by squaring positive quantities. Second-layer caps on distinct owners of the same label have product at most $(3/4)8 + 8 = 14$; distinct labels give at most $(3/4)16 = 12$. The two caps of one owner have product $(3/4)32 - 8 = 16$. A second-layer cap and a pure tail have product at most $\sqrt{8}\sqrt{32} = 16$. All old-old pairs retain their original bounds.

The head squared norms in the equator, first layer, second layer, and pure-tail layer are respectively 32, $64/3$, 24, and zero, so different layers cannot coincide. Within a layer, distinct heads and tail labels give distinct points. Each of the 311 new owners is removed once and inserted twice. Hence

$$196560 - 2090 - 311 + 3(2258) + 2(311) + 12 = 201567.$$

Dividing all points by $\sqrt{32}$ proves the unit-sphere claim. $\square$

The finite optimization behind the second layer has a particularly small geometric description. Keep only shell vectors that are not first-layer owners and satisfy (4.14). Connect two such vectors when their product exceeds eight. One seeks a largest subset that can be split into three independent sets of this graph. Assign each independent set to one coordinate line. Distinct-label products are already controlled by the Leech-shell bound. The published 310-owner witness and the new 311-owner witness are feasible assignments in this same model; each extra owner contributes one net point.

5. LAYERED BINARY CODES AND JOINT SUPPORT EXCHANGES

We now keep the layer geometry fixed and improve its binary ingredients, using replacements whose cost is the union of their old conflicts.

5.1. Compatible replacements and shared deletion costs. Let $X \subset S^{d-1}$ be a kissing configuration and let $C \subset S^{d-1} \setminus X$ be a finite candidate set. For $c \in C$, put $F(c) = \{x \in X : \langle c, x \rangle > 1/2\}$, and join two candidates in a graph when their inner product exceeds $1/2$. For $I \subseteq C$, write $F(I) = \bigcup_{c \in I} F(c)$.

Proposition 5.1 (Shared-deletion replacement). *For $D \subseteq X$ and $I \subseteq C$, the set $(X \setminus D) \cup I$ is a kissing configuration if and only if I is independent and $F(I) \subseteq D$. In particular, for independent I , the largest configuration with inserted set I retains $X \setminus F(I)$ and has net gain*

$$X_I = (X \setminus F(I)) \cup I, \quad |X_I| - |X| = |I| - |F(I)|. \quad (5.1)$$

More generally, for a positive weight function w on $X \cup C$, the greatest weight gain with inserted set I is $w(I) - w(F(I))$, where $w(A) = \sum_{a \in A} w(a)$.

Proof. Old-old pairs are compatible; new-new pairs require independence of I ; old-new pairs require $F(I) \subseteq D$. Disjointness of C and X gives the count, and positive weights penalize additional deletions. $\square$

The same count applies to disjoint bundles of points. Partition the deletable old points into nonempty groups X_j , and let Q_i be pairwise disjoint, internally compatible candidate bundles outside X . Require each Q_i to be compatible with the old points held fixed. Join two bundles when some cross pair is incompatible, and let $F(i)$ index the old groups containing a conflict with

Q_i . Put $a_i = |Q_i|$ and $b_j = |X_j|$. Among replacements that retain or delete each old group as a whole, the exact optimization problem is

$$\max \left(\sum_i a_i z_i - \sum_j b_j r_j \right), \quad z_i + z_{i'} \leq 1 \ (ii' \in E), \quad r_j \geq z_i \ (j \in F(i)), \quad z_i, r_j \in \{0, 1\}. \quad (5.2)$$

At an optimum only conflicting groups are removed, each charged once. Single-point groups recover Proposition 5.1; internally compatible pools admit the following exact matching certificate, an application of the classical deficiency form of Hall's theorem [31] to the bipartite conflict graph.

Proposition 5.2 (Optimal subexchange in a compatible pool). *Suppose the entire candidate set C is internally compatible. Let H be the bipartite conflict graph on C and X , with $cx \in E(H)$ exactly when $x \in F(c)$, and let $\nu(H)$ be its maximum matching size. Then*

$$\max_{I \subseteq C} (|I| - |F(I)|) = |C| - \nu(H). \quad (5.3)$$

An optimal I is the set of candidate vertices reachable from the unmatched candidates by alternating paths of a maximum matching. In particular no positive subexchange exists if and only if a matching assigns every candidate a distinct old conflict. The same formula, multiplied by the common bundle size, applies to internally compatible candidates and old groups of equal cardinality.

Proof. A matching meets at most $|F(I)|$ candidates in I , so $\nu(H) \leq |F(I)| + |C \setminus I|$, giving the upper bound in (5.3).

From the unmatched candidates of a maximum matching, traverse unmatched edges toward X and matched edges toward C . For the reached sets $I \subseteq C$ and $J \subseteq X$, every edge out of I reaches J or is the matched edge by which its endpoint in I was reached; hence $J = F(I)$. No vertex of J is unmatched, since that would give an augmenting path. Matching edges biject J with the matched vertices of I , leaving exactly $|C| - \nu(H)$ unmatched vertices in I . This proves equality. $\square$

Internal compatibility of the pool is essential: otherwise an alternating-path set can contain conflicting new candidates, and the matching deficiency is only an upper bound. The certificate concerns the specified candidate pool, not all possible replacements.

5.2. The three-layer construction. Let $n \geq m \geq 32$. Choose a binary code $T \subset \mathbb{F}_2^m$ with minimum distance at least $\lceil m/4 \rceil$, and a support family $\mathcal{B} \subseteq \binom{[n]}{8}$ such that

$$|B \cap B'| \leq 4 \quad (B \neq B', \ B, B' \in \mathcal{B}). \quad (5.4)$$

For the standard basis $e_0, \dots, e_{n-1}$ of $\mathbb{R}^n$, define

$$X_T = \left\{ \frac{1}{\sqrt{m}} \sum_{i=0}^{m-1} (-1)^{c_i} e_i : c \in T \right\}, \quad (5.5)$$

$$X_M = \left\{ \frac{1}{\sqrt{8}} \sum_{i \in B} \varepsilon_i e_i : B \in \mathcal{B}, \ \varepsilon_i \in \{\pm 1\}, \ \prod_{i \in B} \varepsilon_i = 1 \right\}, \quad (5.6)$$

$$X_R = \left\{ \frac{\varepsilon e_i + \varepsilon' e_j}{\sqrt{2}} : 0 \leq i < j < n, \ \varepsilon, \varepsilon' \in \{\pm 1\} \right\}. \quad (5.7)$$

The final layer consists of the normalized D_n roots.

Proposition 5.3 (Three-layer specialization of ERS [6]). *The union $X_T \cup X_M \cup X_R$ is a kissing configuration with*

$$|X| = |T| + 128|\mathcal{B}| + 2n(n-1). \quad (5.8)$$

Proof. All points have norm one. Support sizes m , eight, and two distinguish the layers, whose sizes are $|T|$, $2^7|\mathcal{B}|$, and $4\binom{n}{2}$.

In the dense layer, two codewords at distance d have inner product $1 - 2d/m \leq 1/2$. On one middle support, two distinct even sign patterns differ in at least two positions, giving at most $(8 - 4)/8 = 1/2$. On different middle supports, (5.4) gives at most four contributions of $1/8$. Two roots with different supports share at most one coordinate, giving at most $1/2$; on the same support distinct roots have inner product zero or -1 .

For the three cross-layer cases, the numbers of overlapping coordinates give respectively

$$\langle X_T, X_M \rangle \leq \sqrt{8/m}, \quad \langle X_T, X_R \rangle \leq \sqrt{2/m}, \quad \langle X_M, X_R \rangle \leq \frac{2}{\sqrt{8}\sqrt{2}} = \frac{1}{2}. \quad (5.9)$$

Here each expression bounds all pairs from the indicated layers, even when a middle support extends beyond the first m coordinates. Since $m \geq 32$, all bounds are at most $1/2$. $\square$

The inputs are uncoupled: one additional support contributes 128 points, and each additional sign word contributes one.

5.3. The support families. For eight-element supports the identity

$$d_H(B, B') = 16 - 2|B \cap B'|$$

identifies (5.4) with minimum distance eight. The following finite objects supply the middle layers.

Theorem 5.4. *The following constant-weight-code lower bounds hold:*

n	33	34	37	38	39
$A(n, 8, 8) \geq$	1803	1960	2843	3077	3383.

Finite verification. In each complete binary-mask list $\mathcal{B}_n$ of Table 9, every mask lies in $[0, 2^n)$, has eight set bits, and occurs once. Pairwise checks give the intersection distributions in Table 3, all at most four. Independently, enumerating the five-subsets gives $56|\mathcal{B}_n|$ distinct keys: a repeated key is equivalent to an intersection of size at least five. These checks establish the codes. $\square$

TABLE 3. Support cardinalities and numbers h_j of unordered pairs with intersection size j . All other intersection counts are zero.

n	$ \mathcal{B}_n $	h_0	h_1	h_2	h_3	h_4
33	1803	134282	389920	680407	234225	185669
34	1960	168266	488949	783677	281186	197742
37	2843	443987	1255368	1461343	621750	257455
38	3077	555513	1521825	1676186	706467	272435
39	3383	718023	1896522	1985532	820689	299887

The final support families arise from exchanges in the constant-weight space. For an incumbent family $\mathcal{B}$ and a new support c , let $F(c) = \{b \in \mathcal{B} : |b \cap c| > 4\}$. If $Y \subseteq \binom{[n]}{8} \setminus \mathcal{B}$ is internally compatible, then

$$\mathcal{B}' = (\mathcal{B} \setminus F(Y)) \cup Y, \quad F(Y) = \bigcup_{c \in Y} F(c), \quad |\mathcal{B}'| - |\mathcal{B}| = |Y| - |F(Y)|. \quad (5.10)$$

This specializes (5.2) to 128-point bundles; shared deletions can make a joint exchange profitable when no single insertion is.

The final transitions are

n	33	34	37	38	39
parent size	1802	1959	2840	3076	3382
$ F(Y) $	9	6	37	6	14
$ Y $	10	7	40	7	15
final size	1803	1960	2843	3077	3383.

The certificates include complete parents, insertions and deletions. The checker recomputes $F(Y)$ and reconstructs each final family by (5.10); the complete witnesses independently prove the theorem.

The conflict graphs of these five candidate pools have maximum matching sizes 9, 6, 37, 6, 14, respectively: in each case all removed supports can be matched to distinct inserted supports. By Proposition 5.2, the best subexchange using only the supplied inserted supports therefore has gain 1, 1, 3, 1, 1. The displayed exchanges attain these values within the supplied pools, not necessarily among all replacements.

5.4. Dense layers from coset unions. Let $H \subset \mathbb{F}_2^m$ be linear and let $r_0, \dots, r_{q-1}$ be coset representatives. Define

$$T = \bigcup_{a=0}^{q-1} (r_a + H), \quad \delta_{ab} = \min_{h \in H} \text{wt}(r_a + r_b + h) \quad (a \neq b). \quad (5.11)$$

Within a coset, distances are the nonzero weights of H ; between cosets, the difference set is $r_a + r_b + H$. Thus the δ_{ab} are their minimum distances, and positivity makes the cosets disjoint, giving $|T| = q2^{\dim H}$.

Proposition 5.5. *The generators and representatives in Appendix B define the sign codes in Table 4.*

TABLE 4. Parameters of the dense sign codes.

m	$\dim H$	$d(H)$	q	$d(T)$	$ T $	Used in dimension
32	17	8	1	8	131072	32–37
38	14	12	11	10	180224	38
39	15	12	10	10	327680	39

Finite verification. Binary elimination gives the three displayed ranks. Enumeration of the spans gives the weight distributions in Appendix B, whose least positive weights are eight, twelve, and twelve. The length-32 code uses the sole representative zero. For lengths 38 and 39, enumeration of (5.11) gives $\delta_{ab} = 10$ for all 55 and 45 distinct representative pairs, respectively. The preceding difference-set argument proves the sizes and minimum distances. Finally, each distance is at least $\lceil m/4 \rceil$, as required by Proposition 5.3. $\square$

The length-32 code is the Cheng–Sloane [32, 17, 8] code [23], in the generator convention of Grassl’s table [32]. The other two kernels have a common quadratic-residue origin. Let

$$g(x) = x^{23} + x^{19} + x^{18} + x^{14} + x^{13} + x^{12} + x^{10} + x^9 \\ + x^7 + x^6 + x^5 + x^3 + x^2 + x + 1 \in \mathbb{F}_2[x]. \quad (5.12)$$

Take the coefficient vectors of $x^i g(x)$, $0 \leq i < 24$, as length-47 binary words, and append their parity at coordinate 47. This specifies the extended quadratic-residue code in the chosen convention [3, Chapter 16]. Shorten at coordinates 40, $\dots$, 47, retaining coordinates 0, $\dots$, 39 in order, to

obtain a dimension-16 code H_{40} . Shortening H_{40} at coordinate zero gives the printed H_{39} ; shortening at coordinates zero and one gives the printed H_{38} . Here shortening imposes zeros before deleting coordinates. Exact row-space comparisons verify these identities and, in the displayed retained-coordinate conventions,

$$H_{38} = \{(h_0, \dots, h_{37}) : (h_0, \dots, h_{37}, 0) \in H_{39}\}. \quad (5.13)$$

The distance proof uses the explicit generators and representatives, not an assumed distance of the mother code.

The five layered-code constructions. For each $n \in \{33, 34, 37, 38, 39\}$, combine the support family of Theorem 5.4 with the sign code of Proposition 5.5. Proposition 5.3 gives Table 5. The dimension-37 and dimension-38 counts are improved further in Sections 6 and 7. $\square$

TABLE 5. The three contributions to the five code-based configurations.

n	$ X_T $	$ X_M $	$ X_R $	Total
33	131072	230784	2112	363968
34	131072	250880	2244	384196
37	131072	363904	2664	497640
38	180224	393856	2812	576892
39	327680	433024	2964	763668

At the first three lengths, the increase relative to the cited comparison is 128 times the increase in support size, respectively 15, 24, and 11. At the last two lengths, both the larger support families and the explicit coset unions enter the final count.

5.5. A new length-32 support family. The same construction applies to a verified family of 1676 eight-subsets of $[32]$. There are 1,403,650 unordered support pairs. Their intersection histogram, in orders zero through four, is

$$(112077, 318935, 601033, 196122, 175483).$$

No larger intersection occurs. Together with the same $[32, 17, 8]$ dense code, Proposition 5.3 gives

$$K(32) \geq 2^{17} + 128(1676) + 2(32)(31) = 347584.$$

The family was obtained from Echols's public 1667-word seed [18] through joint exchanges. The final witness, rather than successful replay of a randomized search, is the finite input to this proof. The 2843-word length-37 family above is also improved to 2845 words. Section 6 uses this larger family and changes the sign patterns, giving a further gain.

6. LOCAL SIGNED REPLACEMENTS

6.1. A compatible replacement region. Scale the ERS layers [6] of Proposition 5.3 to squared norm 32: the $[32, 17, 8]$ code gives 2^{17} vectors $(\pm 1)^{32}$ padded to dimension d ; each eight-subset $B \in \mathcal{B}$ carries 128 fixed-parity sign patterns with entries ± 2 ; the roots have two entries ± 4 . Distinct supports in $\mathcal{B}$ intersect in at most four.

For $d = 35$ and 36 , the archived support families have 2,157 and 2,742 blocks, respectively [17]. The base cardinalities are

$$2^{17} + 128 \cdot 2157 + 4 \binom{35}{2} = 409548, \quad 2^{17} + 128 \cdot 2742 + 4 \binom{36}{2} = 484568.$$

Proposition 6.1 (Exact bundle boundary). *Let $Q \subset \{0, \pm 1\}^d$ consist of distinct weight-eight vectors with $q \cdot q' \leq 4$ for distinct q, q' , and let $\mathcal{D} \subseteq \mathcal{B}$. Deleting the complete middle-layer bundles on $\mathcal{D}$ and adjoining $2Q$ preserves the kissing condition and changes the cardinality by $|Q| - 128|\mathcal{D}|$ if and only if*

$$|\text{supp}(q) \cap B| \leq 4 \quad (q \in Q, B \in \mathcal{B} \setminus \mathcal{D}). \quad (6.1)$$

Proof. Write a retained middle-layer vector as $2s$, and put $r = |\text{supp}(q) \cap B|$, where $B = \text{supp}(s)$. If $r < 8$, signs agreeing with q on the intersection extend to either prescribed parity on B : one coordinate of $B \setminus \text{supp}(q)$ can fix the parity. Thus $\max_s q \cdot s = r$. For $r = 8$, the maximum is eight or six according as q has the bundle's parity or not. If $r > 4$, a maximizing $s = q$ violates the additive count, and any other maximizer violates the kissing condition. This proves necessity, including possible duplicates.

Conversely, (6.1) bounds new-retained middle products by 16; new-new products have the same bound by hypothesis. Top-new and bottom-new products are at most $8 \cdot 2 = 16$ and $2 \cdot 4 \cdot 2 = 16$. New vectors have squared norm 32 and distinct supports from retained middle vectors; coordinate magnitudes distinguish them from the other layers. Unchanged pairs remain compatible, proving the count and sufficiency. $\square$

For prescribed Q , the least whole-bundle deletion set making $2Q$ a compatible, disjoint insertion is

$$F(Q) = \{B \in \mathcal{B} : |B \cap \text{supp}(q)| > 4 \text{ for some } q \in Q\}.$$

Its cost is $128|F(Q)|$, as in Proposition 5.1, regardless of the new signs' parity while retained bundles remain complete.

Corollary 6.2 (Isolated replacement). *Let W contain exactly b blocks of $\mathcal{B}$, and suppose every other block meets W in at most four coordinates. Any internally compatible signed weight-eight code Q on W can replace those b bundles, with gain $|Q| - 128b$.*

Proof. Every new support is contained in W , so (6.1) holds after deleting the b contained blocks. $\square$

This is the weight-eight version of the free-zone replacement principle of Takhanov–Yun [25, Definition 5 and Lemma 4]. The exact criterion also permits non-isolated regions with suitable new supports.

In dimension 35 take $W = \{12, \dots, 22\}$; precisely block 2145 lies inside W . In dimension 36 take $W = \{12, \dots, 23\}$; precisely blocks 2739, 2740 and 2741 lie inside W . Indices are zero-based. The independent checker verifies these facts from the complete support lists, together with all boundary intersections.

6.2. Hadamard transfer.

Lemma 6.3 (Paired Hadamard transfer). *Let $\mathcal{A}$ be a family of four-subsets on an even number m of coordinates, with pairwise intersections at most two. Fix a perfect matching of the coordinates and retain the blocks meeting each matched pair in at most one coordinate. If r blocks survive, there exists a signed weight-eight code of size $16r$ in $\{0, \pm 1\}^m$ whose distinct vectors have inner products at most four. If the paired coordinates are embedded in the ERS configuration, write $P(A)$ for the union of the four pairs met by a retained source block A . The transferred code replaces any b old bundles with gain $16r - 128b$ provided*

$$|P(A) \cap B| \leq 4 \quad \text{for every retained source block } A \text{ and undeleted old block } B. \quad (6.2)$$

Proof. Take all sixteen sign patterns on each retained block and apply $H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ to each matched coordinate pair. An occupied pair's single entry ± 1 becomes two entries ± 1 , so every image has weight eight and support $P(A)$. Original inner products are at most two, by the intersection

bound for different supports and at least one sign change for equal supports. Since $H^\top H = 2I$, products double and distinctness is preserved. Proposition 6.1 gives the replacement claim. $\square$

For dimension 35, an exact-cover construction gives a Steiner quadruple system $S(3, 4, 10)$ with 30 blocks. The matching

$$(0, 1), (2, 3), (4, 5), (6, 7), (8, 9)$$

retains 16 blocks, hence 256 signed weight-eight vectors. One extra zero coordinate embeds them into the eleven-coordinate region W .

The twelve-coordinate source is the 51-block design of Kalbfleisch–Stanton [26, Theorem 7], also used in the local-replacement construction of Takhanov–Yun [25, Section V]. For dimension 36, use two copies of a one-factorization of K_6 . In each six-coordinate half, take the complements of the three edges in one factor, giving three internal four-subsets. For each of the five factors, combine each of its three edges in the first half with each of the three in the second half. There are $3 + 3 + 5 \cdot 9 = 51$ blocks, with intersections at most two. The matching

$$(0, 1), (2, 4), (3, 5), (6, 7), (8, 10), (9, 11)$$

retains 36 blocks, hence 576 signed weight-eight vectors. The same cross-block construction works for every even half-size, and is sharp in its complete-sign, transverse $2 + 2$ model. Let L, R be disjoint sets of $2t$ coordinates, with prescribed perfect matchings M_L, M_R . Write $E_L = \binom{L}{2} \setminus M_L$ and $E_R = \binom{R}{2} \setminus M_R$. Identify a transverse $2 + 2$ support $e \cup f$ with the pair $(e, f) \in E_L \times E_R$.

Proposition 6.4 (Sharp transverse transfer). *Let $t \geq 2$ and $\mathcal{A} \subseteq E_L \times E_R$. Taking all sixteen signs on each source support and applying the paired Hadamard map gives an internally compatible weight-eight code if and only if both*

$$\{f : (e, f) \in \mathcal{A}\} \quad (e \in E_L), \quad \{e : (e, f) \in \mathcal{A}\} \quad (f \in E_R)$$

are matchings on their respective coordinate sets. Consequently

$$|\mathcal{A}| \leq 2t^2(t-1), \quad |Q| \leq 32t^2(t-1). \quad (6.3)$$

Equality holds precisely when every one of these left and right fibres is a perfect matching. For any prescribed M_L, M_R , equality is attainable. In particular, $t = 3$ gives 576 vectors in twelve coordinates.

Proof. Two source supports with different edges in both halves intersect in at most two positions. If their left edges agree, their intersection has size at most two exactly when their right edges are disjoint; the symmetric statement holds on the right. Complete sign bundles make this intersection condition necessary as well as sufficient: signs agreeing on an intersection of size at least three have product at least three, and hence product at least six after transfer. Lemma 6.3 proves sufficiency and the factor sixteen.

Each fibre has at most t edges, and $|E_L| = |E_R| = \binom{2t}{2} - t = 2t(t-1)$. Summing degrees on either side proves the bound, with equality exactly when every fibre is a perfect matching.

For attainment, take one-factorizations of the two complete graphs; these exist for every t [26, p. 137]. Relabel vertices so that factor zero is the prescribed pairing in each half, and use the same $2t - 1$ colour labels. For each nonzero colour include every union of one left edge and one right edge of that colour. Every fibre is a perfect matching, and there are $(2t - 2)t^2$ blocks. For $t = 3$ this is the four-colour construction used above. $\square$

The equality criterion does not classify extremizers as synchronized one-factorizations. The bound requires both the $2 + 2$ restriction and complete source sign bundles; it does not cover arbitrary signed weight-eight codes.

6.3. Boundary-constrained capacity. For fixed old bundles to be deleted, form the bipartite allowed-block graph $\Gamma \subseteq E_L \times E_R$ by retaining exactly the pairs (e, f) for which $P(e \cup f)$ satisfies (6.2). More generally, Γ can encode any prescribed restrictions on the transverse $2+2$ blocks. For each $e \in E_L$, let G_e be the graph on R with edges $\{f : (e, f) \in \Gamma\}$; define H_f on L symmetrically. Denote the matching number of a graph G by $\nu(G)$.

Proposition 6.5 (Boundary matching bounds). *An allowed family $\mathcal{A} \subseteq \Gamma$ is compatible exactly when all its left and right fibres are matchings. If $C_L \subseteq E_L$, $C_R \subseteq E_R$ form a vertex cover of Γ , then*

$$|\mathcal{A}| \leq \sum_{e \in C_L} \nu(G_e) + \sum_{f \in C_R} \nu(H_f). \quad (6.4)$$

In particular one may use either complete vertex class as a cover, or minimize the displayed weighted cover bound. Any vertex cover of G_e or H_f gives a further explicit upper bound on its matching number.

The unrestricted bound $2t^2(t-1)$ is attainable under Γ if and only if Γ contains a subfamily whose fibres on both sides are perfect matchings. Thus even one fibre with matching number less than t obstructs equality. If U is any upper bound obtained from (6.4), deleting b old bundles and using this model has net gain at most $16U - 128b$.

Proof. The compatibility and equality assertions follow from Proposition 6.4. The degree of e in $\mathcal{A}$ is at most $\nu(G_e)$, and similarly for f . Every selected pair has an endpoint in the chosen vertex cover, so summing these degree bounds proves (6.4). A matching uses distinct vertices of any vertex cover, proving the additional local bound. The last claim counts sixteen inserted points per source block and 128 points per deleted bundle. $\square$

Local matching bounds need not be simultaneously attainable: choices in the G_e can violate the H_f conditions. In twelve coordinates, the geometric boundary resolves this obstruction. Each half contains three coordinate pairs. A left pair type selects two of the three left pairs, and a right pair type selects two of the three right pairs. There are three types on each side. Represent the allowed combinations by a bipartite graph H on these two three-element sets. Every edge $a \in E(H)$ allows all sixteen source blocks choosing one coordinate from each of its four prescribed pairs. This is exactly the restriction produced by (6.2), since the image support depends only on the four occupied pairs, not on the chosen input axes.

Put $h = |E(H)|$. Define a bipartite graph D_H with two copies a^-, a^+ of every edge a of H , joining a^- to b^+ when $a \neq b$ share an endpoint in H . Thus D_H is the bipartite double cover of the line graph of H . Matching-vertex-cover duality bounds the capacity; compatible source blocks attain the resulting integer cell weights.

Theorem 6.6 (Exact twelve-coordinate boundary capacity). *The maximum number of compatible transverse $2+2$ source blocks under the type restriction H is*

$$\beta(H) = 8h - 4\nu(D_H) = \max_{M \text{ matching in } H} (8|M| + 4|E(H - V(M))|), \quad (6.5)$$

where $V(M)$ is the set of endpoints of M . Consequently the maximum complete-sign transferred code has $16\beta(H)$ vectors.

Fix a maximum matching N in D_H . For a compatible allowed family, let c_a count its source blocks of type a . It is optimal if and only if

$$c_a + c_b = 8 \quad (a^- b^+ \in N), \quad c_a = 8 \quad \text{if either copy of } a \text{ is unmatched by } N.$$

In particular, an optimum can always be chosen with every c_a equal to zero, four or eight.

Proof. For one type, its four possible left edges each have a right fibre of size at most two, so $c_a \leq 8$. If a, b share a left type, their two distinct right types share a coordinate pair. Every right

edge from these two types meets that pair, so each left fibre has at most two edges among them. Hence $c_a + c_b \leq 8$. The symmetric argument applies when they share a right type.

Apply this last inequality to every edge of N , and $c_a \leq 8$ to each unmatched vertex copy. Each c_a is counted twice, giving

$$2 \sum_a c_a \leq 8|N| + 8(2h - 2|N|) = 16h - 8\nu(D_H).$$

Equality holds exactly when all these inequalities are tight, proving the asserted equality criterion once attainment is established.

Take a minimum vertex cover C of D_H , of size $\nu(D_H)$, and put

$$s_a = 2 - \mathbf{1}_{a \in C} - \mathbf{1}_{a^+ \in C} \in \{0, 1, 2\}.$$

Both directed copies of each adjacency are covered, so $s_a + s_b \leq 2$ whenever a, b share an endpoint in H . Thus weight-two types form a matching in H and have only weight-zero neighbours. To obtain C , follow alternating paths from unmatched minus vertices of N , then take unreached minus and reached plus vertices. Absence of augmenting paths makes this a cover with exactly one endpoint of each matching edge.

For each type with $s_a = 2$, use the eight source blocks whose four binary coordinate choices have even parity. Distinct choices differ in at least two positions, so these blocks intersect in at most two coordinates. For the types with $s_a = 1$, restrict one synchronized 36-block factorization family from Proposition 6.4. That family has four blocks of every type: each of its left fibres is a perfect matching on three coordinate pairs, whose three inter-pair edge counts satisfy $u + v = u + w = v + w = 2$, hence $u = v = w = 1$. The four possible left edges of a type each contribute one block. Weight-one blocks are mutually compatible. A weight-two type and any other positive type differ on both sides, so their blocks intersect in at most one coordinate per half. The combined compatible family has

$$4 \sum_a s_a = 4(2h - |C|) = 8h - 4\nu(D_H)$$

blocks, proving attainment.

More generally, for any matching M in H , take eight parity blocks on each edge of M and four factorization blocks on each edge of $H - V(M)$. The same argument proves compatibility. Conversely, in the constructed optimum the weight-two types form such an M , and the weight-one types lie in $H - V(M)$. These two observations prove the second expression in (6.5). $\square$

Thus a fixed twelve-coordinate pairing, a division into two halves, and b prescribed old-bundle deletions have exact best gain in this model

$$16\beta(H) - 128b = 64(2h - \nu(D_H) - 2b).$$

The joint type constraints can be stronger than the fibre-cover bound. For example, take the five types $H = \{(1, 1), (1, 2), (1, 3), (2, 1), (3, 1)\}$. The two fibre sums are 28 and the best mixed cover gives 24, whereas D_H has a matching of size five and the exact capacity is 20. The function `solve_type_boundary` in `constructions/codes/hadamard/boundary.py` takes the allowed types and returns the exact capacity, an attaining source family, and matching and vertex-cover certificates. The independent tests check all 2^9 type relations. Restrictions on individual source blocks within a type require the finer boundary model of Proposition 6.5.

Corollary 6.7 (One forbidden pair type in twelve coordinates). *Let $t = 3$, so that each half consists of three prescribed coordinate pairs. A pair type specifies two of these pairs in each half, and allows the source edges joining the chosen pairs. If the boundary forbids at least one such type, at most 32 source blocks, hence 512 signed points, are possible. If it forbids exactly one type and allows all the others, these bounds are attained.*

Proof. There are four left edges joining the two left pairs in the forbidden type. For each, all permitted right edges meet the third right pair, a two-vertex cover; its fibre has size at most two instead of three. The other eight left fibres have size at most three, giving $4 \cdot 2 + 8 \cdot 3 = 32$.

The factorization family has exactly four blocks of each pair type, as in the preceding proof. If only one type is forbidden, deleting its four blocks leaves 32 compatible, allowed blocks. $\square$

Proposition 6.8 (Sign-extension bound). *Let $Q \subset \{0, \pm 1\}^m$ have constant weight w and distinct-pair inner products at most s . If $2w - m > s$, then $|Q| \leq 2^w$. In particular, the maximum size of a weight-eight code in eleven coordinates with inner products at most four is 256.*

Proof. A full sign string $\sigma \in \{\pm 1\}^m$ cannot extend two members of Q : their supports would intersect in at least $2w - m > s$ coordinates, and their signs agree wherever they intersect. Each member has 2^{m-w} extensions. Hence $2^{m-w}|Q| \leq 2^m$. The 256-point transferred code above attains the bound for $(m, w, s) = (11, 8, 4)$. $\square$

Theorem 6.9. *There are kissing configurations of sizes 409,676 in dimension 35 and 484,760 in dimension 36.*

Proof. The two replacement sizes give

$$N_{35} = 2^{17} + 128(2157 - 1) + 256 + 4 \binom{35}{2} = 409676,$$

$$N_{36} = 2^{17} + 128(2742 - 3) + 576 + 4 \binom{36}{2} = 484760.$$

Proposition 5.3 controls the unchanged pairs; Corollary 6.2 and Lemma 6.3 control the replacements. The archived data verify the support, region and transverse-block hypotheses. $\square$

6.4. A 512-point replacement in dimension 37. Proposition 6.1 only tests the selected image supports, rather than their entire containing region. Take the verified 2845-word length-37 family and delete three complete sign bundles. The replacement consists of 512 signed weight-eight vectors on twelve coordinates. In the source's one-based, left-to-right numbering, the paired coordinates are

$$(18, 20), (23, 24), (15, 21), (16, 22), (14, 17), (19, 25).$$

Thirty-two transverse four-subsets on the twelve input axes, each with all sixteen signs, give the 512 vectors by the preceding Hadamard transfer. The four-subsets, pairs and signed images in `constructions/codes/data/d37_signed_patch.json` regenerate exactly the supplied 512-point set. The checker verifies internal products at most four and intersections at most four with every retained support. The boundary-adapted transfer therefore gives

$$K(37) \geq 2^{17} + 128(2845 - 3) + 512 + 2(37)(36) = 498024.$$

Against the 2832-support comparison family, the thirteen extra supports contribute $13 \cdot 128 = 1664$ points. The signed replacement contributes $512 - 3 \cdot 128 = 128$ further points, for a total gain of 1792.

For a sharp application of Corollary 6.7, label the six displayed coordinate pairs $P_0, \dots, P_5$ in order. Of the fifteen possible four-pair image supports, exactly twelve meet every retained old support in at most four coordinates. The forbidden three are those whose two omitted pair indices are

$$\{0, 2\}, \quad \{0, 5\}, \quad \{2, 4\}.$$

These three edges form the path 4, 2, 0, 5. Every division of the six pairs into two triples cuts at least one edge of this path, because its four vertices cannot all lie in a single triple. Such a crossing edge is the omitted pair of a forbidden $2 + 2$ type, so every division has capacity at

most 512. The division $\{0, 2, 4\} \mid \{1, 3, 5\}$ cuts exactly one edge. Deleting the four blocks of its forbidden type from a 36-block factorization construction therefore supplies an alternative boundary-compatible 512-point code. Consequently 512 is the exact capacity among complete-sign $2 + 2$ transfers, allowing any division into two triples but keeping these six coordinate pairs and these three deleted bundles fixed. More precisely, applying Theorem 6.6 to the ten divisions, identifying left–right reversals, gives four capacities of 512, four of 448 and two of 384. The finite checker constructs an attaining family for every division and verifies each transferred code against the complete retained boundary.

The supplied 32-block witness itself fits no such division: its image supports use ten different four-pair types, whereas a fixed $2 + 2$ division permits only nine. It belongs to the more general model of Lemma 6.3. Neither the restricted capacity statement nor the 576-point isolated construction in dimension 36 proves optimality for arbitrary twelve-coordinate signed codes.

Left-to-right coordinate i corresponds to integer bit $37 - i$ (zero-based), or coordinate $38 - i$ numbered from the right. The verifier uses this conversion consistently for supports and signed vectors.

7. SECOND-SHELL COMPLETION IN DIMENSION 38

The preceding layered-code construction gives 576892 points in dimension 38. The larger bound in Table 1 starts from the public 591612-point configuration [15]. Its decomposition has an unoccupied equator. This permits the addition of directions from a different Leech shell.

Lemma 7.1 (Equatorial completion). *Let $q, b > 0$. Suppose $X \subset S^{D+k-1}$ is a kissing configuration consisting of points $(u/\sqrt{q}, t)$ and pure-tail points $(0, p)$. Let $V \subset \mathbb{R}^D$ be a set of representatives of distinct antipodal lines, all of squared norm b , such that*

$$|\langle v, v' \rangle| \leq b/2 \quad (v, v' \in V, v \neq v'), \quad (7.1)$$

$$|\langle v, u \rangle| \leq \sqrt{bq}/2 \quad (v \in V, (u/\sqrt{q}, t) \in X). \quad (7.2)$$

Assume every t in a point with nonzero head is nonzero. Then adjoining all $(\pm v/\sqrt{b}, 0)$ gives a kissing configuration with $|X| + 2|V|$ points.

Proof. New points have norm one. Their products with pure tails vanish, with caps are $\pm \langle v, u \rangle / \sqrt{bq}$, and with other new lines are $\pm \langle v, v' \rangle / b$; opposite points have product -1 . The nonzero tails separate old and new points. $\square$

7.1. The base configuration. Use the integer Golay–Leech shell Σ of squared norm 32, with the Golay generator stored in the dimension-38 package. Partition its 98280 antipodal lines into classes with pairwise absolute inner products at most eight. The tail data consist of 1932 vectors of squared norm eight in $\mathbb{R}^{14}$, partitioned into 644 zero-sum triples. Inside a triple the mutual products are -4 ; between any distinct tail vectors they are at most four.

Each mother line is assigned a triple. It contributes six points of the form

$$(\varepsilon u / \sqrt{48}, d / \sqrt{24}), \quad \varepsilon \in \{-1, 1\}, \quad (7.3)$$

where d runs through its assigned triple. Their squared norm is $32/48 + 8/24 = 1$. Pairs in one class at one tail have product at most $8/48 + 8/24 = 1/2$. Repeated heads at different tails in their triple have product $32/48 - 4/24 = 1/2$. Different heads and different tails have product at most $16/48 + 4/24 = 1/2$. Opposite heads give smaller products.

There are also 1932 pure-tail points. Write the tail vectors as the rows of an integer matrix F , and the stored rational rotation as M/h , with M integral and $h > 0$. Its defining checks are

$$MM^t = h^2 I_{14}, \quad |(FM^t F^t)_{ij}|^2 \leq 48h^2 \quad \text{for all } i, j. \quad (7.4)$$

The pure-tail points are $(0, dM^t/(h\sqrt{8}))$. Orthogonality preserves their norms and mutual products. Their products with the caps in (7.3) are entries of $FM^tF^t/(h\sqrt{192})$, so the second inequality gives absolute products at most $1/2$. Thus both required checks use exact integer arithmetic. The layer counts are

$$\underbrace{0}_{\text{equator}} + \underbrace{6(98280)}_{\text{caps}} + \underbrace{1932}_{\text{pure tails}} = 591612.$$

The class partition, triples, rotation, and Golay generator are inherited data from the public construction; the completion below is the additional part.

7.2. The 144 added lines.

Lemma 7.2 (Cross-shell compatibility). *Let $\mathcal{L}$ be a lattice of minimum squared norm m . If $u, v \in \mathcal{L}$ have distinct squared norms a, b , then $|u \cdot v| \leq (a + b - m)/2$.*

Proof. The distinct norms imply $u \pm v \neq 0$. Applying the minimum bound to these two lattice vectors gives $a + b \pm 2u \cdot v \geq m$. $\square$

Corollary 7.3 (An automatic shell window). *In Lemma 7.1, suppose all nonzero heads u are minimal vectors of one lattice of minimum squared norm m , and $q > m$. For any shell norm b with $m < b \leq q$, every lattice vector v of squared norm b satisfies (7.2). Hence any set of lines from that shell satisfying (7.1) gives an equatorial completion.*

Proof. Lemma 7.2 gives $|v \cdot u| \leq b/2$, so each new-old product is at most $\frac{1}{2}\sqrt{b/q} \leq 1/2$. It is strictly below $1/2$ when $b < q$. The head norm $\sqrt{m/q} < 1$ ensures nonzero cap tails. $\square$

In the integer Leech model, $m = 32$ and $q = b = 48$ use the endpoint of this window, giving $|v \cdot u| \leq 24$ against every minimal vector. The new certificate lists 144 integer rows v of squared norm 48. They are distinct modulo sign and satisfy

$$|v \cdot v'| \leq 24 \quad (v \neq v'), \quad \max_{u \in \Sigma} |v \cdot u| \leq 24. \quad (7.5)$$

Each row belongs to the same Leech lattice: for some $\varepsilon \in \{0, 1\}$ all coordinates have parity ε , $((v_i - \varepsilon)/2 \bmod 2)_i$ is a Golay word, and $\sum_i v_i \equiv 4\varepsilon \pmod{8}$. Thus only mutual products constrain selection. The checker verifies membership and recomputes cross products independently. Both arrays use the base construction's Golay basis.

Proposition 7.4 (Contacts at the shell-window endpoint). *Let $\mathcal{L}$ have minimum squared norm m , and let $v \in \mathcal{L}$ have squared norm b , where $m < b < 4m$. At $q = b$, a cap with a minimal head u is in contact with the equatorial point $v/\sqrt{b}$ exactly when u belongs to*

$$C_v = \{u \in \mathcal{L} : u^2 = (v - u)^2 = m\}.$$

They are paired without fixed points by $u \mapsto v - u$. For the Leech lattice with $(m, b) = (32, 48)$, every such C_v has 552 elements. Each added point in the dimension-38 construction therefore has exactly 1656 contacts with the old cap layer.

Proof. A minimal head is in contact precisely when $v \cdot u = b/2$, equivalently $(v - u)^2 = m$. The map is an involution; a fixed point would give $v = 2u$ and $b = 4m$.

Rescale to minimum four. The Leech minimal shell L is an 11-design [33, Theorem 4.1], so for $v^2 = s$,

$$M_{2j}(s) := \sum_{x \in L} (v \cdot x)^{2j} = \frac{196560(4s)^j(2j-1)!!}{24 \cdot 26 \cdots (24+2j-2)} \quad (1 \leq j \leq 5). \quad (7.6)$$

For $s = 6$, integrality and the minimum give $|v \cdot x| \leq 3$. The polynomial $a^2(a^2 - 1)(a^2 - 4)$ isolates the extremes; antipodality gives

$$|C_v| = \frac{M_6(6) - 5M_4(6) + 4M_2(6)}{720} = 552.$$

Each oriented minimal head in (7.3) occurs at three tails, giving $3(552) = 1656$ old cap contacts. Pure-tail points are orthogonal to the added equatorial point. $\square$

The contact checker verifies the count and involution in integers for all 144 stored lines.

Theorem 7.5. *There is a kissing configuration of 591900 points in dimension 38.*

Proof. Apply Lemma 7.1 with $q = b = 48$ to the base configuration (7.3) and the 144-line witness. Its hypotheses are exactly (7.5). The new points have zero tail, while every old cap has tail norm $1/\sqrt{3}$; the old pure-tail points have zero head. Thus the count is $591612 + 2(144) = 591900$. $\square$

The norm-48 search produced a 142-line witness and then the 144-line witness used here. All minimal directions occur in the caps, but they do not exhaust the admissible equatorial directions.

7.3. Multiple shells and the next-shell obstruction. The automatic window controls new-old pairs, not pairs between different added shells. The distinction has an exact formulation.

Proposition 7.6 (Multi-shell completion). *Under the hypotheses of Corollary 7.3, let $V_i \subset \mathcal{L}$ represent distinct antipodal lines of squared norm b_i , where $m < b_i \leq q$. Adjoining $(\pm v/\sqrt{b_i}, 0)$ for every $v \in V_i$ gives $|X| + 2 \sum_i |V_i|$ distinct kissing points if and only if*

$$|v \cdot w| \leq \frac{\sqrt{b_i b_j}}{2} \iff \min\{\|v - w\|^2, \|v + w\|^2\} \geq b_i + b_j - \sqrt{b_i b_j} \quad (7.7)$$

for every two distinct labelled representatives $(v, i), (w, j)$.

Proof. All new-old products satisfy the automatic shell bound. Products between the two signs of one new line equal -1 , while the largest product between two labelled lines is $|v \cdot w|/\sqrt{b_i b_j}$. This proves necessity, sufficiency and distinctness. The equivalence follows from $\|v \pm w\|^2 = b_i + b_j \pm 2v \cdot w$. $\square$

The ambient minimum bound is weaker than the required separation: for $b, c > m$,

$$b + c - \sqrt{bc} \geq \sqrt{bc} > m.$$

In the current Leech normalization, evenness makes every squared norm a multiple of 16. The automatic interval $32 < b \leq 48$ therefore contains only the norm-48 shell. The next shell is excluded more strongly:

Proposition 7.7 (No norm-64 equatorial addition). *Every norm-64 Leech vector has exactly 46 minimal vectors u with $v \cdot u = 32$, and 46 with $v \cdot u = -32$. Neither $v/8$ nor $-v/8$ can be adjoined at the equator of the unchanged base configuration (7.3).*

Proof. Rescale to minimum four and $v^2 = 8$. For $a_x = v \cdot x$ on the minimal shell, integrality and the minimum give $|a_x| \leq 4$. Using (7.6) and the extreme-value polynomial $P(a) = a^2(a^2 - 1)(a^2 - 4)(a^2 - 9)$ gives

$$|\{x : a_x = 4\}| = \frac{M_8(8) - 14M_6(8) + 49M_4(8) - 36M_2(8)}{40320} = 46.$$

The negative count is the same. Returning to integer coordinates, each of these positive heads gives the forbidden product $32/\sqrt{64 \cdot 48} = 1/\sqrt{3} > 1/2$ with $v/8$. Only this extreme positive product violates the bound. Every oriented minimal head appears at three tails, so each proposed endpoint has exactly 138 conflicts with the old cap layer. $\square$

8. SEVEN LIFTS FROM A COMPATIBLE LINE CLASS

One compatible P_{48p} line class gives seven antipodal block lifts. Their gains depend on both the class size and the union of its selected images.

8.1. The tail configurations. For dimensions $k = 1, \dots, 7$, use respectively the normalized root configurations $A_1, A_2, D_3, D_4, D_5, E_6, E_7$. They have 2, 6, 12, 24, 40, 72, 126 points and are antipodal. Explicitly, $A_1 = \{-1, 1\}$, and A_2 is the six points $(e_i - e_j)/\sqrt{2}$ in the plane $x_1 + x_2 + x_3 = 0$. The D_k roots are $(\varepsilon e_i + \varepsilon' e_j)/\sqrt{2}$, with $i < j$. Their kissing conditions follow directly from their coordinate overlaps.

The last two codes come from the integer E_8 model

$$W = \{\pm 2e_i \pm 2e_j : i < j\} \cup \{(\varepsilon_0, \dots, \varepsilon_7) : \varepsilon_i \in \{\pm 1\}, \prod_i \varepsilon_i = 1\}. \quad (8.1)$$

Coordinate overlaps and even sign distance give squared norm eight and distinct-row products at most four. The subsets

$$W_7 = \{w \in W : w_6 + w_7 = 0\}, \quad W_6 = \{w \in W : w_5 = w_6 = -w_7\}$$

have respectively $62 + 64 = 126$ and $40 + 32 = 72$ rows. The maps

$$w \mapsto (w_0, \dots, w_5, \sqrt{2}w_6), \quad w \mapsto (w_0, \dots, w_4, \sqrt{3}w_5)$$

are isometries on the indicated subspaces. Dividing by $\sqrt{8}$ therefore gives the required 7- and 6-dimensional antipodal codes.

8.2. Rigidity of antipodal lifts with pure tails. Pure tails fix the lifting geometry even when paired heights may differ and depend on the head.

Proposition 8.1 (Paired-lift rigidity). *Let $C \subset S^{D-1}$ and $Y \subset S^{k-1}$ be kissing configurations, with Y containing distinct antipodal pairs $\{\pm u_i\}$, $1 \leq i \leq t$. Choose $B_i \subseteq C$ and, for each $x \in B_i$, heights $0 < h_{i,x}^+, h_{i,x}^- < 1$. Retain $C \setminus \bigcup_i B_i$ at the equator, retain all of Y as pure tails, and insert*

$$\left(\sqrt{1 - (h_{i,x}^\pm)^2} x, \pm h_{i,x}^\pm u_i \right).$$

The resulting set is a kissing configuration if and only if every height is $1/2$, the sets B_i are pairwise disjoint, and distinct heads in each B_i have inner product at most $1/3$.

Proof. The pure tail of the same sign requires $h_{i,x}^\pm \leq 1/2$. Write these two heights as $\sin \theta_+$ and $\sin \theta_-$, where $0 < \theta_\pm \leq \pi/6$. The inner product of the two copies of x is $\cos(\theta_+ + \theta_-)$. It is at most $1/2$ only if $\theta_+ + \theta_- \geq \pi/3$, forcing both heights to equal $1/2$.

If the same head belongs to two blocks, choose the signs of its tails to make their product nonnegative. The two corresponding points have inner product $3/4 + |\langle u_i, u_j \rangle|/4 \geq 3/4$, a contradiction. Within one block, the same-tail condition is exactly $(3/4)\langle x, x' \rangle + 1/4 \leq 1/2$. These prove necessity; Theorem 2.5 proves sufficiency. $\square$

Now suppose C is antipodal. For a pool of compatible antipodal line classes $\mathcal{B}_1, \dots, \mathcal{B}_m$ of C , suppose each of $t \leq m$ tail pairs is assigned a subset of one selected class. The largest configuration obtainable in this model has exactly

$$|C| + |Y| + 2 \max_{|I|=t} \left| \bigcup_{i \in I} \mathcal{B}_i \right| \quad (8.2)$$

points: disjointness gives the upper bound, and assigning each line, with both signs, to its first selected class attains it.

8.3. An orbitwise image-union guarantee. Standard averaging and conditional expectations [34] give the following bound without assuming transitivity.

Proposition 8.2 (Orbitwise image union). *Let a finite group $\mathcal{G}$ act on a finite set Ω , with orbits O_j of sizes N_j . For $\mathcal{B} \subseteq \Omega$, put $b_j = |\mathcal{B} \cap O_j|$ and $q_j = 1 - b_j/N_j$. For every integer $t \geq 1$, there are $g_1, \dots, g_t \in \mathcal{G}$ such that*

$$\left| \bigcup_{i=1}^t \mathcal{B}g_i \right| \geq \left\lceil \sum_j N_j (1 - q_j^t) \right\rceil. \quad (8.3)$$

They can be selected deterministically: if U_r is the union after $r < t$ choices, choose $g_{r+1} \in \mathcal{G}$ to maximize

$$\sum_j q_j^{t-r-1} |(\mathcal{B}g_{r+1} \setminus U_r) \cap O_j|. \quad (8.4)$$

Here $q_j^0 = 1$, including when $q_j = 0$.

Proof. Each point of O_j is missed by a uniform image with probability q_j , and by t independent images with probability q_j^t . After r choices the conditional expected union size is

$$\Phi_r = \sum_j [N_j - q_j^{t-r} (N_j - |U_r \cap O_j|)].$$

Rule (8.4) maximizes Φ_{r+1} , whose average over the next image is Φ_r . Hence $|U_t| = \Phi_t \geq \Phi_0$; integrality gives the ceiling. $\square$

For compatible line classes, each covered line contributes two points to the lift. A transitive action gives the expected union $N[1 - (1 - b/N)^t]$ for N lines and a b -line class. The seven bounds below use certified image unions, not this average.

8.4. Selection from a finite image pool. For a directly computable guarantee, let $B_1, \dots, B_m$ be a finite pool of supplied line classes, put $\Omega = \bigcup_i B_i$, and let $a_x = |\{i : x \in B_i\}|$. Distinct indices are selected without replacement; the classes themselves may overlap.

Proposition 8.3 (Finite-pool selection). *For $0 \leq t \leq m$, some t classes have union size at least*

$$\left\lceil \sum_{x \in \Omega} \left(1 - \frac{\binom{m-a_x}{t}}{\binom{m}{t}} \right) \right\rceil. \quad (8.5)$$

Such a selection can be obtained using integer scores. At a stage with M classes remaining and h choices still to make, let U be the current union and a_x the number of remaining classes containing $x \notin U$. Choose a remaining class B_i maximizing

$$\sum_{x \in B_i \setminus U} \binom{M-1-a_x}{h-1}. \quad (8.6)$$

Here a binomial coefficient is zero if its upper index is negative or smaller than its lower index.

Proof. A uniform t -subset of the m indices misses x with probability $\binom{m-a_x}{t} / \binom{m}{t}$, giving (8.5) by averaging. Conditional on the choices already made, the expected final size after choosing i is

$$|U| + \frac{1}{\binom{M-1}{h-1}} \sum_{x \notin U \cup B_i} \binom{M-1-a_x}{h-1}.$$

Thus maximizing (8.6) maximizes this conditional expectation. Points in every remaining class contribute equally for every choice and are covered at this step; the stated zero convention handles them. The average over the next choice is the current expectation, so it never decreases. At the final step it equals the actual union size, proving the bound. $\square$

The implementation in `constructions/p48/coverage.py` collects equal incidence sets $\{i : x \in B_i\}$ with their multiplicities. Its subsequent exchanges use exact union gains, preserving the guarantee.

8.5. The 7077-line class and its automorphism images. We use the catalogue lattice P_{48p} in its published integral Gram basis G [29]. Its minimum squared norm is six. Since it is even unimodular of rank 48, the theta-series calculation in (C.6) gives 52416000 minimal vectors. The witnesses use this basis directly, not the quadratic-residue neighbour of Section 10.1.

Represent a minimal line by an integer row c satisfying $cGc^t = 6$, identifying c and $-c$. The new mother class $\mathcal{C}$ has 7077 lines and satisfies

$$|cGc^t| \leq 2 \quad (c \neq \pm c', \ c, c' \in \mathcal{C}). \quad (8.7)$$

Every norm and pair product in this assertion is an integer. The certificate supplies the complete class and matrices A_i with $A_iGA_i^t = G$. Thus $\mathcal{C}A_i$ is another class with the same property.

For t selected images, assign each line to the first image in which it occurs. Write $\mathcal{D}_i$ for the assigned subset of $\mathcal{C}A_i$, and put

$$S = \left| \bigcup_{i=1}^t \mathcal{C}A_i \right| = \sum_{i=1}^t |\mathcal{D}_i|.$$

The $\mathcal{D}_i$ are disjoint as line sets. Taking both signs of each assigned line gives disjoint vector blocks B_i of size $2|\mathcal{D}_i|$, with normalized inner products at most $1/3$ by (8.7). Assign these blocks injectively to the t distinct antipodal lines of the corresponding root tail code. Theorem 2.5 gives

$$K(48 + k) \geq 52416000 + 2S + \tau_k, \quad \tau_k = 2t. \quad (8.8)$$

Theorem 8.4. *The class and image matrices in the supplementary data give the seven bounds in Table 6.*

Proof. The finite checker verifies the catalogue Gram matrix, its exact isometries, all 7077 norms, distinctness modulo sign, and all distinct-line inequalities (8.7). It then reconstructs each selected image and counts its union with the preceding images. This gives the S values in the table. The root tail codes have the ranks, sizes, and inner products established above. Hence (8.8) applies. $\square$

TABLE 6. The line-class liftings. The union, rather than the sum of image sizes, determines the number of available lines.

d	t	S	τ_k	$52416000 + 2S + \tau_k$
49	1	7077	2	52430156
50	3	21231	6	52458468
51	6	42459	12	52500930
52	12	84874	24	52585772
53	20	141328	40	52698696
54	36	253851	72	52923774
55	63	442507	126	53301140

8.6. Class growth and image selection. The public comparison uses a 7069-line class $\mathcal{C}_0 \subset \mathcal{C}$ [15]. With the selected matrices held fixed, let

$$S_0 = \left| \bigcup_{i=1}^t \mathcal{C}_0 A_i \right|,$$

and let S_{ref} be the union size in the comparison construction. The increase in the final point count splits exactly as

$$2(S - S_{\text{ref}}) = 2(S - S_0) + 2(S_0 - S_{\text{ref}}). \quad (8.9)$$

The terms measure class enlargement with images fixed and image selection with the original class fixed, respectively.

TABLE 7. The two contributions to the seven point-count improvements.

d	S_0	$2(S - S_0)$	$2(S_0 - S_{\text{ref}})$	Total
49	7069	16	0	16
50	21207	48	2	50
51	42411	96	18	114
52	84778	192	64	256
53	141168	320	154	474
54	253565	572	296	868
55	442009	996	414	1410

The three selected images in dimension 50 are disjoint. Larger image families overlap, so an eight-line enlargement need not supply eight new lines per image. In dimension 55 it contributes 498 distinct lines, rather than $8 \cdot 63 = 504$. The image choice contributes a further 207 lines.

9. EMBEDDED ROOT SYSTEMS AND MOMENT-CERTIFIED SECTIONS

An embedded root system imposes restrictions on the projections of the entire minimal shell. In the case below, these restrictions determine the number of points perpendicular to a child section.

Theorem 9.1 (E_8 -extension count). *Let L be an extremal even unimodular lattice of rank 48, and let $\iota : \sqrt{3}E_8 \hookrightarrow L$ be an isometric embedding. Use the simple-root numbering in (9.8), and set*

$$U = \text{span}_{\mathbb{R}}\{\iota(\sqrt{3}\alpha_1), \dots, \iota(\sqrt{3}\alpha_5)\}.$$

Then exactly 2,553,792 vectors of squared norm 6 in L are perpendicular to U . The same count holds for every $W(E_8)$ -conjugate of this D_5 subsystem inside the embedded E_8 .

The proof uses integrality, the lattice minimum and spherical-design moments, not an extension of the Weyl group to automorphisms of L .

9.1. Signature domains and target statistics. The standard design identities [33, 35] give the following principle. The signature domain must be complete, and averaging must preserve the target statistic.

Proposition 9.2 (Finite-section statistics). *Let $L \subset \mathbb{R}^n$ be an integral lattice of minimum squared norm μ , and suppose its minimal shell X has N points and is a spherical $(2s+1)$ -design. Choose independent $b_1, \dots, b_r \in L$ with Gram matrix H . For $x \in X$ set*

$$y(x) = (\langle x, b_i \rangle)_{i=1}^r, \quad q(y) = y^t H^{-1} y, \quad f(y) = |\{x \in X : y(x) = y\}|.$$

Choose a finite antipodal set $\mathcal{R} \subset \mathbb{Z}^r \setminus \{0\}$ of section-vector coefficients, and put

$$\begin{aligned} E &= \{Hz : z \in \mathcal{R}, z^t Hz = \mu\}, \\ D &= \{y \in \mathbb{Z}^r : q(y) \leq \mu, |z^t y| \leq z^t Hz/2 \text{ for every } z \in \mathcal{R}\}. \end{aligned} \quad (9.1)$$

Then every signature lies in the disjoint union $D \cup E$, and $f(e) = 1$ for each $e \in E$. No primitivity of the section is required.

For a multi-index α of total degree at most $2s + 1$, define $M_\alpha = 0$ when $|\alpha|$ is odd, and, when $|\alpha| = 2m$, define

$$M_\alpha = \frac{N\mu^m}{\prod_{j=0}^{m-1} (n + 2j)} \sum_P \prod_{(a,b) \in P} H_{i_a i_b}, \quad (9.2)$$

where the labelled list $(i_1, \dots, i_{2m})$ contains α_i copies of i , and P runs over its pairings. Empty products give $M_0 = N$.

Let Γ be a finite group of linear transformations preserving q , D and E , and let $\psi : D \cup E \rightarrow \mathbb{R}$ be Γ -invariant. For its orbits O_j on D , form

$$A_{\alpha j} = \sum_{y \in O_j} y^\alpha, \quad b_\alpha = M_\alpha - \sum_{e \in E} e^\alpha, \quad c_j = \sum_{y \in O_j} \psi(y).$$

Any selected moment rows in the stated degree range may be used. If $A^t \lambda \leq c$ entrywise, then

$$\sum_{x \in X} \psi(y(x)) \geq \sum_{e \in E} \psi(e) + \lambda^t b. \quad (9.3)$$

If $A^t \lambda = c$, equality holds.

Proof. Integrality and orthogonal projection give $y(x) \in \mathbb{Z}^r$ and $q(y(x)) \leq \mu$. For $a = \sum_i z_i b_i$ and $x \neq \pm a$, $\|x \pm a\|^2 \geq \mu$ gives $|z^t y(x)| \leq \|a\|^2/2$. The exceptions have $\|a\|^2 = \mu$: their full norm lies in the section, so each fibre is a singleton and violates its own domain inequality. Thus $D \cap E = \emptyset$; positive definiteness of q makes D finite.

Polarization gives (9.2). Since every isometry of q extends orthogonally to $\mathbb{R}^n$, these moments are Γ -invariant even when Γ does not act on L . Hence

$$\bar{f}(y) = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} f(g^{-1}y).$$

is nonnegative, has the same moments and equals one on E . Its orbit values satisfy $Av = b$, while invariance of ψ preserves $\sum_X \psi = \sum_E \psi + c^t v$. The result follows from $(c - A^t \lambda)^t v \geq 0$. $\square$

Antipodality removes odd total-degree rows, not monomials with odd individual exponents. Exceptional orbits may instead be retained with value one. A noninvariant target is only averaged, not determined for a fixed child. In dimension 45, longer section vectors remove all columns carrying slack in a one-sided certificate.

9.2. A realization in P_{48n} . Take $L = P_{48n}$, whose Gram matrix G is published as CQ48a in the Nebe–Sloane catalogue [28]. It is an extremal even unimodular lattice of rank 48, minimum squared norm 6, with minimal shell

$$X = \{x \in L : \|x\|^2 = 6\}, \quad |X| = 52,416,000.$$

The parent certificate [19] specifies eight integer coefficient rows T and scaled coordinate rows B , satisfying

$$TGT^t = H = 3C(E_8) = BB^t/96. \quad (9.4)$$

The catalogue and certificate Grams agree entry by entry. The first five rows T_5 have Gram

$$H_5 = \begin{pmatrix} 6 & 0 & -3 & 0 & 0 \\ 0 & 6 & 0 & -3 & 0 \\ -3 & 0 & 6 & -3 & 0 \\ 0 & -3 & -3 & 6 & -3 \\ 0 & 0 & 0 & -3 & 6 \end{pmatrix}, \quad \det H_5 = 972. \quad (9.5)$$

This is $3C(D_5)$ in the stated numbering. Both embeddings are primitive (all nonzero Smith invariants are one), though the theorem does not require this. For $U = \text{span}_{\mathbb{R}}(T_5)$, $\dim U^\perp = 43$.

Corollary 9.3. *The set $Y = X \cap U^\perp$ has 2,553,792 elements. Consequently the ordinary Euclidean kissing number satisfies $K(43) \geq 2,553,792$.*

Theorem 9.1 gives the cardinality. Since $\|x - x'\|^2 \geq 6$ for distinct shell vectors, $Y/\sqrt{6}$ is a kissing configuration.

The construction is specified without a list of millions of points. In integer basis coordinates its defining set is

$$\mathcal{Z} = \{z \in \mathbb{Z}^{48} : zGz^t = 6, T_5Gz^t = 0\}. \quad (9.6)$$

For $EE^t = G$, its points are $zE/\sqrt{6}$ with $z \in \mathcal{Z}$; the certificate supplies G, T_5 .

9.3. The complete projection domain. Now let L be any lattice satisfying Theorem 9.1, and let X be its squared-norm-six shell. The theta-series identity (C.6) gives $|X| = 52,416,000$. Identify the parent section with $\sqrt{3}E_8$, with simple roots $\sqrt{3}\alpha_1, \dots, \sqrt{3}\alpha_8$. Write the orthogonal projection of $x \in X$ onto this section as $p = \lambda/\sqrt{3}$. Integrality of L and self-duality of E_8 imply $\lambda \in E_8$. Set

$$y_i = \langle \lambda, \alpha_i \rangle = \langle x, \sqrt{3}\alpha_i \rangle, \quad q = \|p\|^2 = \|\lambda\|^2/3 = y^t H^{-1} y \leq 6.$$

For an E_8 root r , if $x \neq \pm\sqrt{3}r$, applying the minimum bound to $x \pm \sqrt{3}r$ gives

$$|\langle \lambda, r \rangle| \leq 3. \quad (9.7)$$

The exceptions are $\lambda = 3r$. Each such projection has multiplicity one: its squared length is already 6, so the perpendicular component vanishes.

For a transparent finite enumeration use the model

$$E_8 = \{u/2 : u \in \mathbb{Z}^8, u_i \text{ all of the same parity, } \sum_i u_i \equiv 0 \pmod{4}\}.$$

Its roots are the 112 vectors $\pm e_i \pm e_j$ and the 128 half-integer vectors with entries $\pm 1/2$ and an even number of minus signs. The simple roots used here are

$$\begin{aligned} \alpha_1 &= (1, -1, -1, -1, -1, -1, -1, 1)/2, & \alpha_2 &= e_1 + e_2, \\ \alpha_{i+2} &= -e_i + e_{i+1} & & (1 \leq i \leq 6). \end{aligned} \quad (9.8)$$

Their Gram is precisely $H/3$.

Enumeration of $\|u\|^2 \leq 72$ subject to (9.7) gives

$\ \lambda\ ^2$	0	2	4	6	8
Nonexceptional signatures	1	240	2,160	6,720	17,280

There are no other nonexceptional signatures. Including the 240 exceptions gives 26,641 signatures in total.

For completeness, pair roots bound the two largest absolute coordinates of u by a sum of 6. Half-integer roots bound $\|u\|_1$ by 12, except when all coordinates are nonzero with odd sign parity, when they bound $\|u\|_1 - 2 \min_i |u_i|$ by 12. Thus $\|u\|_1 \leq 18$ is a safe enumeration bound. Exact parity, norm and root tests give the table; at norm eight they remove the 240 doubled roots from 17,520 points.

9.4. Moments and the averaging explanation. Venkov's theorem [33, Theorem 4.1] makes X an 11-design, so (9.2) applies through degree ten. The abstract Weyl group has five nonexceptional orbits, with sizes s_j and projected squared norms $q_j = 0, 2/3, 4/3, 2, 8/3$. Their averaged multiplicities a_j satisfy the radial equations

$$\sum_j s_j a_j q_j^k = |X| 6^k \frac{(4)_k}{(24)_k} - 240 6^k, \quad 0 \leq k \leq 5, \quad (9.9)$$

where $(a)_k$ is the rising factorial and the subtraction removes the 240 section roots. The first five equations give

$$(a_0, a_1, a_2, a_3, a_4) = (1, 092, 000, 116, 235, 9, 180, 495, 63/4).$$

The equation for $k = 5$ also holds.

The roots perpendicular to the first five simple roots form A_3 . The target subspace contains respectively 1, 12, 6, 24, 0 admissible signatures at the five norms, and 12 exceptional roots. Thus the average contact count over the Weyl images of this D_5 child is

$$1, 092, 000 + 12(116, 235) + 6(9, 180) + 24(495) + 12 = 2, 553, 792. \quad (9.10)$$

This proves an average count, not yet the count of a fixed child. The a_j need not be integers because they are orbit averages.

9.5. Exact count for the specified child. For the fixed child use the target-preserving subgroup

$$\Gamma = \langle W(D_5), W(A_3), -1 \rangle.$$

This group preserves the full domain and the condition $y_1 = \dots = y_5 = 0$, and has 43 orbits O_j . For averaged multiplicities v_j , use rows

$$A_{\alpha,j} = \sum_{y \in O_j} y^\alpha,$$

with right-hand side (9.2), and fix exceptional orbit values to one. The target is

$$c_j = |\{y \in O_j : y_1 = \dots = y_5 = 0\}|.$$

The supplied certificate selects 43 moment or fixed-root rows, and provides a rational column vector w such that

$$A^t w = c, \quad w^t b = 2, 553, 792. \quad (9.11)$$

Thus $|X \cap U^\perp| = c^t v = w^t b$. All inputs depend only on the root system and spherical moments. Precomposing ι with a Weyl element proves the conjugate cases without extending it to L . The verifier regenerates A, b, c from the coordinate model, taking only the row selection and w from the certificate in Table 9.

9.6. A nested family of fixed coordinate sections. Extend ι linearly and define the eight fixed coordinate cuts

$$U_k = \iota(\sqrt{3} \text{span}\{e_1, \dots, e_k\}), \quad 1 \leq k \leq 8.$$

For $k \geq 2$ these are the coordinate D_k spans ($D_2 = A_1^2$, $D_3 = A_3$); U_1 is an axis, not a D_1 root subsystem, and U_8 is the whole parent.

Theorem 9.4 (Coordinate-section family). *Let L, X, ι satisfy Theorem 9.1. For $m = 8 - k$, the number of minimal vectors perpendicular to the fixed space U_k is*

$$\begin{aligned} |X \cap U_k^\perp| = & 1092000 + 18360m + 464944 \binom{m}{2} + 11880 \binom{m}{3} \\ & + 146880 \binom{m}{4} + 2520 \binom{m}{5} + 31680 \binom{m}{6}. \end{aligned} \quad (9.12)$$

The same holds for every Weyl conjugate of this nested family. In particular, a Weyl conjugate contains the specified D_5 child of Theorem 9.1 as its rank-five member.

Proof. Use the doubled coordinates $u = 2\lambda$ of the complete nonexceptional domain D above. Let Γ_k consist of coordinate permutations within the first k and last $8 - k$ positions and sign changes in an even number of positions. This subgroup of $W(D_8)$ preserves D and the target $u_1 = \dots = u_k = 0$. Its orbits are specified by the two multisets of squared coordinates and $\prod_i u_i$. The product distinguishes sign parity when no coordinate is zero; a zero coordinate removes that restriction.

For a partition a , write m_a for the monomial symmetric polynomial whose distinct monomials have coefficient one. Use all rows

$$F_{a,b}(u) = m_a(u_1^2, \dots, u_k^2) m_b(u_{k+1}^2, \dots, u_8^2), \quad |a| + |b| \leq 5,$$

where the partition lengths are at most k and $8 - k$, respectively. For orbit masses w_j set $A_{(a,b),j} = F_{a,b}(u_j)$, with $u_j \in O_j$. Subtracting the 240 singleton exceptions gives $Aw = b$ for the actual masses, without assuming fibre symmetry. The functionals u_i have Gram $12I_8$, so for $|a| + |b| = d$ the unadjusted moment is

$$\frac{52416000 \cdot 72^d}{48 \cdot 50 \cdots (48 + 2d - 2)} \nu_k(a) \nu_{8-k}(b) \prod_i (2a_i - 1)!! \prod_j (2b_j - 1)!!, \quad (9.13)$$

where $\nu_l(a)$ is the number of distinct permutations of a padded with zeros to length l .

All eight matrices have full column rank: regenerated square minors have the nonzero residues in Table 8. Thus w is unique over $\mathbb{Q}$. The candidate

$$w_j = |O_j| a_r, \quad (a_0, a_2, a_4, a_6, a_8) = (1092000, 116235, 9180, 495, 63/4), \quad r = \|u_j\|^2/4,$$

comes from the full Weyl average and satisfies every row exactly. It must therefore equal the actual orbit masses. This does not assign the fractional value $63/4$ to individual fibres.

It remains to count the target signatures. Since $k \geq 1$, a vector λ perpendicular to U_k has integer coordinates and even coordinate sum in the remaining $m \leq 7$ positions. At norms 0, 2, 4, 6, 8, the numbers of admissible signatures are respectively

$$1, \quad 4 \binom{m}{2}, \quad 2m + 16 \binom{m}{4}, \quad 24 \binom{m}{3} + 64 \binom{m}{6}, \quad 160 \binom{m}{5}.$$

These come from the signed shapes 0, (1^2) , (2) , (1^4) , $(2, 1^2)$, (1^6) and $(2, 1^4)$. The root bound excludes $(2, 2)$, and (1^8) cannot fit in $m \leq 7$ coordinates. There are also $4 \binom{m}{2}$ exceptional roots. Multiplication by a_r and addition of the exceptions proves (9.12). Full column rank, not radial moments alone, establishes these counts for the fixed cuts.

For the specified D_5 correspondence put $H_4 = J_4 - 2I_4$ and let $P_{5,8}$ exchange coordinates five and eight. The map

$$Q = \frac{1}{2} P_{5,8} \text{diag}(H_4, H_4)$$

permutes the E_8 roots; the certificate expresses it as 31 simple reflections. It sends $\alpha_1, \dots, \alpha_5$ to

$$-e_1 - e_5, \quad e_3 + e_4, \quad e_1 - e_2, \quad e_2 - e_3, \quad e_3 - e_4,$$

which span the first five coordinate directions. Thus Q^{-1} sends the rank-five cut to the specified D_5 . No ambient Weyl action is required. $\square$

9.7. What accounts for the improvement. The comparison in [19] has 2,545,056 contacts from a primitive $\sqrt{3}D_5$ in P_{48p} . Both sections have determinant 972, singleton cluster, $\beta = 0$ and permutation-isometric Grams. The gain therefore lies in the embedding: the specified E_8 parent in P_{48n} forces the larger perpendicular shell, despite identical intrinsic section data.

TABLE 8. Exact fixed coordinate-section counts and rank certificates. The minors use every orbit column and the specified subset of moment rows.

k	Moment rows	Orbit columns	Minor modulo 1000003	$ X \cap U_k^\perp $
1	45	25	910777	16815624
2	62	34	790744	10663920
3	70	39	894205	6688960
4	72	41	314211	4149504
5	70	39	401115	2553792
6	62	34	244188	1593664
7	45	25	296071	1110360
8	19	13	272057	1092000

10. A MOMENT-CERTIFIED PROJECTION IN DIMENSION 45

Projecting three fibres of a quadratic-residue neighbour gives 7380090 points. A design-moment identity expresses their total size through one fibre, which an anchor graph counts exactly.

10.1. The specified quadratic-residue neighbour. Let C be the extended ternary quadratic-residue code of length 48. We use its published parameters [48, 24, 15] [36, 37]. The supplied generator fixes a degree-23 factor of $1 + x + \cdots + x^{46}$ over $\mathbb{F}_3$, its coordinate order and a self-dual extension coordinate. Set

$$M = \{z \in \mathbb{Z}^{48} : z \bmod 3 \in C\}, \quad L = M/\sqrt{3}, \quad L_0 = \{z/\sqrt{3} \in L : z^2 \equiv 0 \pmod{6}\}.$$

An integer basis B_0 for M and a characteristic numerator $\chi \in M$ are supplied. They satisfy

$$\chi_i \text{ odd for all } i, \quad \chi^2 = 264, \quad \chi \cdot z \equiv z^2 \pmod{6} \quad (z \in M).$$

The last congruence is checked on B_0 and extends by integrality. Define

$$\Lambda = L_0 \cup (\chi/(2\sqrt{3}) + L_0).$$

The characteristic relation and $\chi^2/12 = 22$ make Λ even and integral. The two index-two constructions give covolume one. Integer row operations on B_0 and the two cosets reconstruct a basis B with $BB^t = 12G$, where G is the Gram of the 298-vector certificate; the verification compares all entries.

Lemma 10.1. *The specified lattice Λ has minimum squared norm six.*

Proof. In L_0 , a putative vector of squared norm two or four has numerator z with $z^2 \leq 12$. The minimum code weight forces $z \bmod 3 = 0$. Its nonzero squared norm is then $3w^2$, whose positive even values are at least six.

In the other coset, the numerator $y = \chi + 2z$ has all coordinates odd, so $y^2/12 \geq 4$. Equality requires $y_i = \pm 1$. Let $s_i \in \{-1, 1\}$ represent $\chi_i \bmod 3$. The certificate checks that $s \in C$ and $\chi \cdot s = 66$. Multiplying coordinates by s yields an equivalent extremal ternary code containing the all-one word. Its binary-valued words have weights 0, 24, 48, by the two-symbol specialization of the complete weight enumerator [38, equation (20)]. Write $y = s(1 - 2b)$ coordinatewise. Since $y \bmod 3 \in C$, the binary word b belongs to the normalized code, and its weight w is one of these three values. But $\chi_i s_i \equiv 1 \pmod{6}$, so

$$\chi \cdot z = \frac{\chi \cdot s - \chi^2}{2} - \sum_i \chi_i s_i b_i \equiv -99 - w \equiv 3 \pmod{6}.$$

This contradicts $z/\sqrt{3} \in L_0$. Evenness excludes other positive squared norms below six, and $\sqrt{3}(e_0 + e_1) \in L_0$ has squared norm six. $\square$

Thus the specified Gram has a minimal shell of size 52416000 by (C.6), and this shell is an 11-design [33].

10.2. A complete signature domain and its moments. For the counting argument let Λ be any extremal even unimodular lattice of rank 48, and let X be its squared-norm-six shell. Thus $|X| = 52416000$ by (C.6), and X is an 11-design. Suppose $t_1, t_2, t_3 \in \Lambda$ have Gram matrix

$$H = 8I_3 - 2J_3 = \begin{pmatrix} 6 & -2 & -2 \\ -2 & 6 & -2 \\ -2 & -2 & 6 \end{pmatrix}, \quad H^{-1} = (I_3 + J_3)/8, \quad \det H = 128. \quad (10.1)$$

For $x \in X$, put $y_i = \langle x, t_i \rangle$ and write $f(y)$ for the number of vectors with that signature. No primitivity hypothesis is needed. The vectors $t_1, t_2, t_3, -t_1 - t_2 - t_3$ form a regular tetrahedron of squared radius six. They and their negatives give the eight exceptional signatures

$$E = \{\pm H e_1, \pm H e_2, \pm H e_3, \pm H(1, 1, 1)^t\},$$

each of multiplicity one. Applying the minimum to these eight vectors, as in Proposition 9.2, puts every other signature in

$$D_0 = \{y \in \mathbb{Z}^3 : |y_i| \leq 3, |y_1 + y_2 + y_3| \leq 3\}.$$

Writing $y_4 = -y_1 - y_2 - y_3$ gives $q(y) = \sum_{i=1}^4 y_i^2/8 \leq 9/2$. There are $\binom{15}{3} - 4\binom{8}{3} = 231$ such quadruples, so $D_0 \cup E$ is the original 239-signature domain.

Each $t_i + t_j$ has squared norm eight, so the minimum applied to $x \pm (t_i + t_j) \neq 0$ also gives

$$|y_i + y_j| \leq 4 \quad (1 \leq i < j \leq 3). \quad (10.2)$$

These inequalities exclude exactly the $6 + 24 = 30$ permutations of $(3, 3, -3, -3)$ and $(3, 2, -3, -2)$ from D_0 . The refined domain D has 201 signatures and 101 antipodal orbits; all eight exceptions remain.

Apply (9.2) to all even total degrees at most ten and subtract E . On D_0 this gives $Av = b$ with 161 rows and 116 variables $v_{[y]} = f(y) = f(-y)$. Its entries are $A_{\alpha, [y]} = (2 - \mathbf{1}_{y=0})y^\alpha$, including monomials with odd individual exponents. The system depends only on H .

Let c be the coefficient vector of

$$f(e_1) + f(e_2) + f(e_3) - 9f(-2, -2, 3).$$

The stored rational multiplier λ satisfies

$$c - A^t \lambda \geq 0, \quad \lambda^t b = 7377408. \quad (10.3)$$

All seven nonzero slack entries belong to the 15 antipodal orbits excluded by (10.2). Restricting to D therefore gives

$$A_D^t \lambda = c_D, \quad \lambda^t b = 7377408.$$

The excluded fibres vanish, so the original lower certificate sharpens to the exact identity

$$f(e_1) + f(e_2) + f(e_3) = 7377408 + 9f(-2, -2, 3). \quad (10.4)$$

The coefficient vector uses 1 on $[e_i]$ and -9 on $[(-2, -2, 3)]$, equivalently half these weights at each antipode in Proposition 9.2. The checker regenerates both domains and all rational identities; individual fibre sizes need not be determined.

10.3. Witness amplification.

Theorem 10.2. *Let Λ be an extremal even unimodular lattice of rank 48, and let $t_1, t_2, t_3 \in \Lambda$ have Gram matrix (10.1). With f as above, projection onto $\text{span}_{\mathbb{R}}\{t_1, t_2, t_3\}^\perp$ and normalization send the three fibres e_1, e_2, e_3 injectively to a kissing configuration with exactly $7377408 + 9f(-2, -2, 3)$ points. In particular, m distinct vectors of squared norm six and signature $(-2, -2, 3)$ prove*

$$K(45) \geq 7377408 + 9m.$$

Proof. Each projected squared norm is $6 - 1/4 = 23/4$. Two original distinct vectors have inner product at most three. Their projected products are therefore at most $3 - 1/4 = 11/4$ within one fibre and $3 - 1/8 = 23/8$ in different fibres. This also excludes collisions. Normalization gives a kissing configuration in dimension 45, whose cardinality is (10.4). Finally $f(-2, -2, 3) \geq m$ for any such witness set. $\square$

10.4. A regular anchor graph. For $p \in \Lambda$ of squared norm eight, define $A_p = \{x \in X : \langle x, p \rangle = 4\}$. The involution $x \mapsto p - x$ pairs its elements. Let $\mathcal{G}_p$ have these pairs as vertices, with two distinct vertices adjacent when representatives have inner product one or three.

Proposition 10.3. *For every extremal even unimodular Λ of rank 48 and every squared-norm-eight $p \in \Lambda$, the graph $\mathcal{G}_p$ is well defined, has 1128 vertices and is regular of degree 368. For nonadjacent distinct vertices represented by r, s , the section*

$$(t_1, t_2, t_3) = (-s, s - p, p - r)$$

has Gram matrix (10.1), and its entire fibre satisfies

$$f(-2, -2, 3) = |N(\bar{r}) \setminus N(\bar{s})| = 368 - |N(\bar{r}) \cap N(\bar{s})|. \quad (10.5)$$

Conversely every ordered section with this Gram matrix has such an anchor representation.

Proof. Integrality and the minimum bound give $|\langle x, p \rangle| \leq 4$ for $x \in X$. For $P(t) = t^2(t^2 - 1)(t^2 - 4)(t^2 - 9)$, the even moments through degree eight give

$$2P(4)|A_p| = 90961920, \quad P(4) = 20160, \quad |A_p| = 2256.$$

The partner $p - x \neq x$ again belongs to A_p . For endpoints r, a from distinct pairs, the minimum gives $\langle r, a \rangle \leq 3$ and $4 - \langle r, a \rangle \leq 3$, hence products in $\{1, 2, 3\}$. Exchanging an endpoint replaces the product by four minus itself, proving well-definedness and the vertex count.

To obtain the degree and the frame identity below, take any $z \in p^\perp$. The mixed design moments are

$$M_k(z) := \sum_{x \in X} \langle p, x \rangle^{2k} \langle z, x \rangle^2 = \frac{52416000 6^{k+1} 8^k (2k-1)!!}{48 \cdot 50 \cdots (48+2k)} \|z\|^2 \quad (1 \leq k \leq 4).$$

Consequently

$$\sum_{x \in A_p} \langle z, x \rangle^2 = \frac{M_4(z) - 14M_3(z) + 49M_2(z) - 36M_1(z)}{2P(4)} = 192\|z\|^2. \quad (10.6)$$

The degree is at most ten. For $u = r - p/2$, $u^2 = 4$ and $\langle u, x \rangle = \langle r, x \rangle - 2$. In (10.6), $r, p - r$ contribute 32, each adjacent pair contributes two and the others zero. Thus $768 = 32 + 2 \deg(\bar{r})$.

For a nonadjacent pair $\langle r, s \rangle = 2$, which gives the asserted Gram matrix directly. Each vertex in $N(\bar{r}) \setminus N(\bar{s})$ has a unique endpoint a with $\langle r, a \rangle = 3$; it satisfies $\langle s, a \rangle = 2$. The vector $w = p - a$ has squared norm six and signature $(-2, -2, 3)$. Conversely that signature forces $\langle s, w \rangle = 2$, $\langle p, w \rangle = 4$ and $\langle r, w \rangle = 1$. Thus $a = p - w$ recovers the unique graph vertex, proving the bijection.

Conversely, from any ordered section with Gram (10.1), take $p = -t_1 - t_2$, $s = -t_1$, and $r = -t_1 - t_2 - t_3$. Their products are $p^2 = 8$, $r^2 = s^2 = 6$, $\langle p, r \rangle = \langle p, s \rangle = 4$ and $\langle r, s \rangle = 2$, giving the required representation. $\square$

For a fixed anchor and the specified Gram, minimizing the codegree therefore maximizes the exact projected count

$$7377408 + 9(368 - |N(\bar{r}) \cap N(\bar{s})|) = 7380720 - 9|N(\bar{r}) \cap N(\bar{s})|.$$

10.5. Signed spectrum and lattice restrictions. Choose one endpoint r_i from each of the 1128 pairs and set $u_i = r_i - p/2$. Let $S_{ii} = 0$ and $S_{ij} = \langle u_i, u_j \rangle$ for $i \neq j$, so $S_{ij} \in \{0, \pm 1\}$ and the unsigned adjacency matrix is $A = (|S_{ij}|)$.

Proposition 10.4. *Under the hypotheses of Proposition 10.3,*

$$\sum_{i=1}^{1128} u_i \otimes u_i = 96I_{p^\perp}, \quad S^2 = 88S + 368I, \quad \text{Spec}(S) = \{92^{(47)}, (-4)^{(1081)}\}. \quad (10.7)$$

For distinct i, j , write $c_{ij} = |N(i) \cap N(j)|$ and let $c_{ij}^\pm$ count common neighbours k with $S_{ik}S_{kj} = \pm 1$. Then

$$c_{ij}^\pm = \frac{c_{ij} \pm 88S_{ij}}{2}. \quad (10.8)$$

In particular, all codegrees are even, nonadjacent pairs have equally many common neighbours of either sign, and adjacent pairs have $c_{ij} \geq 88$.

The unsigned matrix satisfies $A + 16I \succeq 0$. Moreover, no four of the u_i can be signed so that all six mutual products are -1 . Changing the chosen endpoints conjugates S by a diagonal sign matrix; it preserves its spectrum, the unsigned graph and this four-vertex obstruction.

Proof. Dividing (10.6) by two gives $\sum_i \langle z, u_i \rangle^2 = 96\|z\|^2$ for $z \in p^\perp$, and polarization proves the frame identity. Hence the Gram matrix $K = 4I + S$ has rank 47 and satisfies $K^2 = 96K$, proving the quadratic identity and spectrum. For $i \neq j$, the entry $(S^2)_{ij}$ is $c_{ij}^+ - c_{ij}^- = 88S_{ij}$, which proves (10.8).

The Gram matrix of the rank-one tensors $u_i \otimes u_i$, with their usual inner product, has entries $\langle u_i, u_j \rangle^2$ and is $16I + A$. This proves the unsigned spectral bound, not a two-valued spectrum for A . If four signed vectors $\varepsilon_i u_i$ had all mutual products -1 , their sum would be a lattice vector:

$$v = \sum_{i=1}^4 \varepsilon_i r_i - \frac{\sum_{i=1}^4 \varepsilon_i}{2} p \in \Lambda, \quad v^2 = 4 \cdot 4 - 2 \binom{4}{2} = 4.$$

This contradicts the lattice minimum. Finally, replacing r_i by $p - r_i$ negates u_i , giving the asserted diagonal conjugation. $\square$

For every section represented by a nonadjacent pair,

$$N = 7380720 - 9c_{ij} \equiv 0 \pmod{18}. \quad (10.9)$$

Improvement on the count at $c_{ij} = 70$ requires $c_{ij} \leq 68$ and hence $N \geq 7380108$, together with the lattice restrictions above.

10.6. The certified anchor and its optimum. The earlier Pless-lattice route found 292 witnesses. The 298-vector construction instead uses an anchor in the odd-coordinate sector of the QR neighbour: in numerator coordinates it has shape $(3^6, 1^{42})$ up to signs and permutation. The parent certificate specifies the anchor, chosen endpoints and 2256 distinct valid elements of A_p ; the moment count proves that this list is complete. Exact checking of all 428076 unordered nonadjacent pairs gives codegrees 70 to 148, with a unique pair attaining 70. It recovers the stored section T and complete 298-vector fibre. Thus 7380090 is optimal for this fixed anchor and section Gram, not over other anchors, lattices or Grams. Integer checks also verify the frame and signed-graph identities and, independently,

$$TGT^t = H, \quad wGw^t = 6, \quad TGw^t = (-2, -2, 3)^t$$

for the 298 distinct rows w . The supplied $WGT^t = I_3$ certifies primitivity of this section, although the results do not require it.

Corollary 10.5. *The specified QR-neighbour construction gives a kissing configuration in dimension 45 with exactly 7380090 points. In particular, $K(45) \geq 7380090$.*

Proof. Lemma 10.1 establishes the ambient hypotheses. The complete parent graph and Proposition 10.3 give $f(-2, -2, 3) = 368 - 70 = 298$. Theorem 10.2 therefore gives $7377408 + 9(298) = 7380090$ points. $\square$

11. FINITE SEARCH AND MATHEMATICAL STRUCTURE

The constructions expose different freedoms in a constrained configuration: moving a contact layer, changing signed supports, using another lattice shell, or selecting less-overlapping images. Their limits are also explicit. Two cap changes are necessary for the fixed insertion in dimension 25; the next Leech shell cannot augment the unchanged dimension-38 base; and pure tails force the heights and disjoint head assignments in the paired lifts.

The section counts use a different reduction. Moments constrain all admissible projection signatures at once. The embedded parent forces the dimension-43 count, whereas the dimension-45 section vectors remove every signature with positive dual slack, yielding the exact identity (10.4). These arguments identify the required statistic without determining every fibre. Together with the finite witnesses, they yield the nineteen stated lower bounds.

12. FINITE VERIFICATION

Each lower bound has two parts: a mathematical reduction covering all types of point pairs, and a finite witness satisfying the reduction's hypotheses. The supplement contains these witnesses and programs for checking the required identities and inequalities. Construction data, verification programs and mathematical research accounts are available in the project repository:

<https://github.com/Oxelra-AI/Qiushi-Engine-Kissing-Number-Research>.

The fixed supplementary-materials snapshot contains the data and programs used in this article. Table 9 identifies the finite objects used in the proofs; paths are relative to `constructions/`.

TABLE 9. Finite inputs to the construction proofs.

Dimensions	Mathematical input	Location
25	Head data, the 552-point layer and two repair vectors	<code>d25/</code>
27	Five 496-point blocks; first-layer heads and 311 labelled second-layer owners	<code>leech/</code> , <code>d27/</code>
32–34, 37, 39	Complete weight-eight support lists, parent families and exchanges	<code>codes/data/</code>
35–37	Signed local codes and their boundary embeddings	<code>codes/</code>
38	Shell partition, tail rotation and 144 added lines	<code>d38/</code>
49–55	The 7077-line class, isometries and tail assignments	<code>p48/</code>
43	Parent embedding, orbit moments and fixed-cut rank certificates	<code>sections/d43/</code>
45	QR-neighbour basis, moment multipliers, parent graph and 298 witnesses	<code>sections/qr/</code> , <code>sections/d45/</code>

For dimension 25, the checker first certifies the 197579-point input from its integer and rational head data. The 552-point motion has symbolically preserved equalities, as proved in Section 3. The remaining strict inequalities use 192-bit outward-rounded Arb intervals or integer comparisons

after squaring positive quantities. The two repair vectors are given as integer rows with exact normalization.

For the triangular Leech lifting, the inputs are five 496-point blocks, the Golay coordinate convention, and exact tail coordinates. Integer products, membership, disjointness and radical identities establish the hypotheses of the lifting theorem. The separate implementations are `leech/verify.py` and `leech/verify_independent.py`. The updated dimension-27 construction additionally checks all first-layer owners, the 311 second-layer owners and labels, and every interlayer inequality in Section 4.4. For the layered constructions, support intersection checks are combined with the distances of the dense binary code or its coset unions. The five supplied parent families also reproduce the support exchanges, including the exact union of deleted conflicts. The signed replacements additionally check every new signed pair and every boundary support intersection; in dimension 37 the checker regenerates all 512 vectors from the 32 transverse four-subsets.

For the bounds in dimensions 49–55, the checker verifies the 7077-line mother class in the catalogue Gram basis, the exact image matrices, the seven union counts, and an injective assignment to antipodal tail lines. The published minimum and the theta-series identity supply the mother-shell premise. The same images applied to the original 7069-line class verify every term in Table 7. The ambient minimum supplies the cross-block inequality.

For dimension 38, the checker regenerates the Leech shell from the base package’s Golay generator, checks the class partition, tail triples and rational rotation, and verifies the 144 new norm-48 lines against both the complete shell and one another. All pair products in the additional tests are integers. Two implementations check the 144-line extension.

The dimension-43 verifier checks the concrete lattice embedding, then regenerates the 26641-signature domain, subgroup action, moment matrix and target vector from standard E_8 coordinates. It checks the supplied rational multiplier against this regenerated system. These latter steps use only the hypotheses of Theorem 9.1. The coordinate-family checker additionally constructs all eight fixed-cut moment systems and their full-rank minors. Independent integer determinants and the explicit Weyl map check the rank certificates and the correspondence with the original D_5 . The dimension-45 checks reconstruct the specified QR neighbour, verify its exact basis Gram, establish the norm-four parity obstruction, and check all 298 fibre witnesses. The parent-graph calculation checks all 2256 endpoints, their pairing about $p/2$, the universal degree from the tenth moments, and equality between the complete target fibre and the 298 witnesses. It also verifies the signed-frame identity and enumerates all 428076 nonadjacent pairs to determine the optimum for this fixed anchor. A separate rational calculation regenerates the 161-row moment system and checks the supplied multiplier in (10.3) directly. The norm-eight restrictions reduce the 239-signature domain to 209 signatures; on its 101 nonexceptional antipodal orbits the multiplier has zero slack. The independent standard-library checker `sections/d45/verify_statistics.py` reconstructs this equality and the anchor moments over the rationals.

12.1. Autonomous mathematical research. The nineteen constructions arose from Qiushi Engine’s autonomous mathematical research, encompassing construction searches, mathematical analysis and computational verification. Its research connected changes of representation with new finite objects: coordinated layer motion (Section 3), joint label selection (Section 4.4), signed-design transfer (Section 6), and moment functionals adapted to the section counts (Sections 9 and 10). The accompanying research accounts describe these mathematical developments and identify their inputs and resulting constructions. Background on the research system is given in [13].

APPENDIX A. THE INTEGER LEECH SHELL AND TAIL CONVENTIONS

This appendix fixes the normalization of the classical Golay description of the Leech shell [2]. Let $G \subset \mathbb{F}_2^{24}$ be generated by the twelve words in Table 10, with coordinate zero at the left. Enumeration and binary elimination give

$$\dim G = 12, \quad G \subseteq G^\perp, \quad W_G(z) = 1 + 759z^8 + 2576z^{12} + 759z^{16} + z^{24}. \quad (\text{A.1})$$

An *octad* is the support of a weight-eight word of G . Two distinct octads intersect in at most four places, because their symmetric difference is a nonzero word of weight at least eight. For $c \in G$ put $\varepsilon_i(c) = (-1)^{c_i}$ and define

$$\Sigma_A = \{4\varepsilon e_i + 4\varepsilon' e_j : 0 \leq i < j < 24, \varepsilon, \varepsilon' \in \{\pm 1\}\}, \quad (\text{A.2})$$

$$\Sigma_B = \left\{ 2 \sum_{i \in O} \eta_i e_i : O \text{ an octad}, \eta_i \in \{\pm 1\}, \prod_{i \in O} \eta_i = 1 \right\}, \quad (\text{A.3})$$

$$\Sigma_C = \left\{ \sum_{i=0}^{23} \varepsilon_i(c) e_i - 4\varepsilon_j(c) e_j : c \in G, 0 \leq j < 24 \right\}. \quad (\text{A.4})$$

Set $\Sigma = \Sigma_A \cup \Sigma_B \cup \Sigma_C$.

Lemma A.1. *This set satisfies (4.1).*

Proof. The absolute coordinate patterns distinguish the three families. Support and signs recover the parameters of the first two. In the third, the unique coordinate of absolute value three identifies j , after which the signs recover c . Thus the cardinalities are

$$|\Sigma_A| = 4 \binom{24}{2} = 1104, \quad |\Sigma_B| = 759 \cdot 128 = 97152, \quad |\Sigma_C| = 4096 \cdot 24 = 98304.$$

They sum to 196560. The respective squared norms are $16 + 16$, $8 \cdot 4$, and $9 + 23$, all equal to 32.

For two A -vectors, different supports give at most one contribution of 16, and equal supports with different signs give at most zero. For A - B pairs there are at most two contributions of eight; for A - C pairs the bound is $4(3 + 1) = 16$. For B - B pairs on different octads, the intersection bound gives at most $4 \cdot 4 = 16$. On the same octad, different even sign patterns differ at least twice, giving at most $4(8 - 4) = 16$.

For a B - C pair, let O be the octad and j the exceptional coordinate of the C -vector. If $j \notin O$, the inner product is at most $2|O| = 16$. Otherwise put $y_i = \eta_i \varepsilon_i(c)$ on O . Self-orthogonality of G and the even sign condition imply $\prod_{i \in O} y_i = 1$, and the inner product is

$$2 \sum_{i \in O} y_i - 8y_j.$$

If $y_j = 1$, this is at most eight. If $y_j = -1$, at least one other y_i is negative, so the sum is at most four and the expression is at most sixteen.

Finally consider C - C pairs with words c, c' and exceptional coordinates j, k . Set $s_i = (-1)^{c_i + c'_i}$ and $S = \sum_i s_i = 24 - 2 \text{wt}(c + c')$. If $j = k$, the inner product is $S + 8s_j$; distinct vectors then require $c \neq c'$, giving $S \leq 8$. If $j \neq k$, it is $S - 4(s_j + s_k)$. This equals sixteen when $c = c'$, and is at most $8 + 8$ otherwise. These six family pairs exhaust all cases. $\square$

TABLE 10. A binary generator for the Golay code in (A.1). Coordinates run from zero to 23 from left to right.

Row	Binary word
0	010111100101000000000001
1	110111001010000000000010
2	100011011101000000000100
3	000110111110000000001000
4	011101011100000000010000
5	111001101001000000100000
6	001001110111000001000000
7	011010001111000010000000
8	101100011011000100000000
9	101111110000001000000000
10	110010110011010000000000
11	111100100110100000000000

For compatibility with the supplied tail arrays, label V by

$$\begin{array}{c|ccc|ccc}
 i & & & & i & & & \\
 \hline
 0 & -1 & 0 & -1 & 6 & -1 & 0 & 1 \\
 1 & 0 & -1 & -1 & 7 & 1 & -1 & 0 \\
 2 & -1 & -1 & 0 & 8 & 0 & -1 & 1 \\
 3 & 0 & 1 & -1 & 9 & 1 & 1 & 0 \\
 4 & -1 & 1 & 0 & 10 & 0 & 1 & 1 \\
 5 & 1 & 0 & -1 & 11 & 1 & 0 & 1.
 \end{array} \tag{A.5}$$

The triangles in (4.6) then have labels $\{0, 7, 10\}$, $\{1, 6, 9\}$, $\{2, 3, 11\}$, and $\{4, 5, 8\}$. For the pure tails set

$$(\sigma(0), \dots, \sigma(11)) = (0, 3, 4, 1, 5, 2, 9, 6, 10, 7, 8, 11), \quad p_a = Rv_{\sigma(a)}/\sqrt{2}.$$

The source rows $\hat{t}_i, \hat{p}_a$ in `constructions/leech/data/tails.json` are related to these coordinates by right multiplication by

$$O = \begin{pmatrix} \sqrt{2}/2 & 0 & \sqrt{2}/2 \\ -\sqrt{6}/6 & \sqrt{6}/3 & \sqrt{6}/6 \\ -\sqrt{3}/3 & -\sqrt{3}/3 & \sqrt{3}/3 \end{pmatrix}, \quad O^\top O = I. \tag{A.6}$$

In row-vector notation, $\hat{t}_i O = v_i^\top/\sqrt{6}$ and $\hat{p}_a O = v_{\sigma(a)}^\top R^\top/\sqrt{2}$. The exact coordinate-change test checks these identities for every label, so the common rotation preserves both within-set and cross-set inner products.

APPENDIX B. BINARY GENERATORS AND COSET REPRESENTATIVES

Table 11 gives the generators used for the three dense sign codes. An entry is a hexadecimal mask, representing the binary word c by $\sum_i c_i 2^i$. The generator of length 32 is used unchanged in ambient dimensions 32–37. The length-38 and length-39 columns are the kernels reconstructed from (5.12). Table 12 supplies the affine representatives; the length-32 code uses only zero.

The complete kernel weight distributions are given in Table 13. The least nonzero weights give the within-coset distance bounds. For every distinct pair of representatives in Table 12, the affine space $r_a + r_b + H_m$ has minimum weight ten. Enumeration of these spaces establishes both the cross-coset distances and disjointness of the cosets. This determines each code in Proposition 5.5 entirely from the printed data.

TABLE 11. Generator masks for H_{32} , H_{38} , and H_{39} . Blank entries reflect their different dimensions.

Row	H_{32}	H_{38}	H_{39}
0	4EF0C081	2000C0DE0B	4000152727
1	24A09082	1000AA229D	2000C0DE0B
2	48609084	8009F5CD6	1000AA229D
3	1E505088	4004FAE6B	8009F5CD6
4	6C905090	200ED9AAD	4004FAE6B
5	FA30C0A0	100BC80CE	200ED9AAD
6	AAC000C0	805E4067	100BC80CE
7	D2608100	40E56DAB	805E4067
8	7E60D200	20B8FB4D	40E56DAB
9	B8601400	1096303E	20B8FB4D
10	D800D800	84B181F	1096303E
11	CC606000	4EFC197	84B181F
12	1EE10000	2BDAD53	4EFC197
13	B2B20000	1949B31	2BDAD53
14	74740000		1949B31
15	D8D80000		
16	FF000000		

TABLE 12. Representatives of the eleven cosets in length 38 and the ten cosets in length 39.

a	r_a in length 38	r_a in length 39
0	0	0
1	C6E94	ADCFB
2	DF6ABF	F7795F
3	27B441	740E29
4	D3CE66	940DFF
5	41BBBB	3B66AB
6	BC1E1F	6CD301
7	52727	4709D4
8	9951C8	98A926
9	7001EB	D23688
10	3F6969	

APPENDIX C. AN EXPLICIT PLESS NEIGHBOUR AND ITS MINIMAL SHELL

This appendix establishes the minimum and shell size in an explicit integer model of an even neighbour. The theta-series argument applies to every extremal even unimodular lattice of rank 48, including the catalogue lattice in Section 8 and the QR neighbour in Section 10.

C.1. The specified even neighbour. Index the rows and columns of a 24×24 matrix S by $\{\infty\} \cup \mathbb{F}_{23}$, in that order. With χ the quadratic character modulo 23, set

$$S_{\infty\infty} = 0, \quad S_{\infty j} = 1, \quad S_{i\infty} = -1, \quad S_{ij} = \chi(i - j) \quad (i, j \in \mathbb{F}_{23}). \quad (\text{C.1})$$

The identity $SS^T = 23I$ follows from the quadratic-character sum $\sum_x \chi(i - x)\chi(j - x) = -1$ for $i \neq j$. Consequently the ternary code

$$C_3 = \{(u, uS) : u \in \mathbb{F}_3^{24}\} \subset \mathbb{F}_3^{48}$$

TABLE 13. Weight distributions $A_w = |\{h \in H_m : \text{wt}(h) = w\}|$. Every omitted weight has coefficient zero.

w	H_{32}	H_{38}	H_{39}
0	1	1	1
8	908	0	0
10	3328	0	0
12	14784	669	970
14	27392	0	0
16	38246	5289	8952
18	27392	0	0
20	14784	8002	16452
22	3328	0	0
24	908	2310	5988
28	0	113	402
32	1	0	3

is self-dual: its systematic generator has rank 24, and $I + SS^T = 24I = 0$ over $\mathbb{F}_3$. Put

$$M = \{z \in \mathbb{Z}^{48} : z \bmod 3 \in C_3\}, \quad L = M/\sqrt{3}. \quad (\text{C.2})$$

The row basis

$$\begin{pmatrix} I_{24} & S \\ 0 & 3I_{24} \end{pmatrix}$$

has determinant 3^{24} . The code orthogonality makes L integral, and the scaling gives covolume one. It is odd, since it contains $3e_0/\sqrt{3}$ of squared norm three.

Let $L_0 = \{x \in L : x \cdot x \in 2\mathbb{Z}\}$ and define

$$c = (1^{24}, -23, 1^{23}), \quad g = c + 6e_0, \quad L' = L_0 \cup \left(\frac{g}{2\sqrt{3}} + L_0 \right). \quad (\text{C.3})$$

The characteristic congruence $c \cdot z/3 \equiv z \cdot z/3 \pmod{2}$ for $z \in M$ is checked on the displayed basis. This suffices since the squared norm modulo two is additive on an integral lattice. Also $g/\sqrt{3} \in L_0$, $g \cdot g/3 = 208$, and $g/(2\sqrt{3})$ has squared norm 52. Its product with $z/\sqrt{3} \in L_0$ is integral by the characteristic congruence. Thus (C.3) is an even lattice, obtained by extending the index-two sublattice L_0 by two cosets, and has covolume one.

The matrix (C.1) is the defining matrix of the Pless symmetry code $C(23)$, with the same signs and coordinate order as [39, Section 1]. In particular, C_3 has minimum Hamming weight 15 [39, 40]. The sum of the rows of $(I \mid S)$ is c , so C_3 contains the all-one word. We shall also use the following consequence of the complete weight enumerator of an extremal ternary self-dual [48, 24, 15] code containing that word:

$$\sum_{b \in C_3 \cap \{0,1\}^{48}} t^{\text{wt}(b)} = 1 + 94t^{24} + t^{48}. \quad (\text{C.4})$$

This is the two-symbol specialization of [38, equation (20)]. These two code-theoretic facts determine which of the possible short vectors can occur in the even neighbour.

Theorem C.1 (The mother shell). *The lattice L' in (C.3) is even unimodular, has minimum squared norm 6, and has exactly 52416000 vectors of squared norm 6.*

Proof. Evenness and unimodularity were established above. Suppose first that $z/\sqrt{3} \in L_0$ has squared norm 2 or 4. Then $z \cdot z$ is 6 or 12. A nonzero residue $z \bmod 3$ would have Hamming weight at least 15, whereas its weight is at most $z \cdot z$. Hence $z = 3w$ for some integer vector w .

Its squared norm is $3w \cdot w$, which is even only when $w \cdot w$ is even, and is therefore at least 6 if nonzero.

A vector in the other coset has the form $y/(2\sqrt{3})$, where $y = g + 2z$ and $z/\sqrt{3} \in L_0$. All 48 coordinates of y are odd, so its squared norm is at least 4. Equality would force $y \in \{1, -1\}^{48}$. Let b be the incidence vector of its negative coordinates and put $w = \text{wt}(b)$. Since $g \equiv \mathbf{1} \pmod{3}$, the identity $z = (y - g)/2$ gives $-z \equiv b \pmod{3}$. Consequently $b \in C_3$, and (C.4) implies $w \in \{0, 24, 48\}$. On the other hand, $c_i \equiv 1 \pmod{6}$, $\sum_i c_i = 24$, and $c \cdot g = 582$. It follows that

$$c \cdot z = \frac{\sum_i c_i - c \cdot g}{2} - \sum_{i:b_i=1} c_i \equiv 3 - w \equiv 3 \pmod{6}. \quad (\text{C.5})$$

This contradicts $z/\sqrt{3} \in L_0$, which requires $c \cdot z \equiv 0 \pmod{6}$. Thus neither coset contains a vector of squared norm 2 or 4. The vector $\sqrt{3}(e_0 + e_1) \in L_0$ has squared norm 6, proving the asserted minimum.

It remains to count the minimum shell. Write $\Theta_{L'}(q) = \sum_{x \in L'} q^{x \cdot x/2}$. The theta series of an even unimodular lattice of rank 48 is a modular form of weight 24 [2, 41]. The space has basis $E_4^6, E_4^3 \Delta, \Delta^2$. Its constant term and the vanishing coefficients of q and q^2 therefore give

$$\Theta_{L'} = E_4^6 - 1440E_4^3 \Delta + 125280\Delta^2. \quad (\text{C.6})$$

Substituting $E_4 = 1 + 240q + 2160q^2 + 6720q^3 + O(q^4)$ and $\Delta = q - 24q^2 + 252q^3 + O(q^4)$ yields

$$\Theta_{L'} = 1 + 52416000q^3 + O(q^4),$$

as required. $\square$

Within this Pless-code model, the shift can equivalently be written using $g_* = \mathbf{1} + 6e_0$: indeed, $(g - g_*)/2 = -12e_{24} \in \sqrt{3}L_0$, so g and g_* define the same coset of L_0 . This is the all-one neighbour construction in [38]; the parity calculation above explains directly why this choice excludes the norm-four vectors. The liftings of Section 8 use the catalogue Gram basis and its published minimum directly; the projection of Section 10 uses the separately specified QR neighbour.

The norm-six shell of L' , divided by $\sqrt{6}$, is a kissing configuration C of size 52416000. Indeed, for two distinct shell vectors x, x' , the nonzero lattice vector $x - x'$ has squared norm at least six, so $x \cdot x' \leq 3$.

APPENDIX D. FLEXIBILITY OF AUXILIARY CONFIGURATIONS

The layer inequalities identify which constraints are tight and which leave room for deformation. A *contact* is a pair of distinct points with inner product $1/2$. Here X_{25} is the 197058-point endpoint model of (4.2), and X_T, X_M, X_R are the ERS layers of Section 5. The dimension-38 observation concerns that layered-code construction; the 591900-point construction instead uses the Leech lift of Section 7. The connection between contact constraints and rigidity of spherical codes is developed in [7]; here it can be made quantitative directly from the proved inner-product margins.

D.1. Paired motion in dimension 25. Write $s \in S_*/\sqrt{32}$ and $x_s^\pm = (\sqrt{3}s/2, \pm 1/2)$, with poles $p^\pm = (0, \pm 1)$. The contacts involving these new points are exactly the 496 paths

$$p^+ - - x_s^+ - - x_s^- - - p^-. \quad (\text{D.1})$$

Their internal vertices are disjoint. They have no contacts with the retained equator, because the relevant inner products are at most $\sqrt{3}/4 < 1/2$. Between different directions the same-latitude bound is $7/16$, and the opposite-latitude bound is $-1/16$. Thus the displayed graph has 994 vertices and 1488 edges.

Proposition D.1. *Keep the equator and the two poles of X_{25} fixed. For each $s \in S_*/\sqrt{32}$ choose independently a unit vector $\hat{s}$ with $\|\hat{s} - s\| < 1/24$. Replacing $x_s^\pm$ by $(\sqrt{3}\hat{s}/2, \pm 1/2)$ preserves the kissing condition and the contacts (D.1).*

Proof. Let $\delta = \max_s \|\hat{s} - s\| < 1/24$. For unit vectors,

$$|\langle \hat{s}, \hat{t} \rangle - \langle s, t \rangle| \leq \|\hat{s} - s\| + \|\hat{t} - t\| \leq 2\delta.$$

Consequently distinct same-latitude directions have inner product less than $7/16 + 3\delta/2 < 1/2$, and distinct opposite-latitude directions have inner product less than $-1/16 + 3\delta/2 < 0$. For a fixed equatorial direction z ,

$$\frac{\sqrt{3}}{2} \langle z, \hat{s} \rangle \leq \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{2} \delta < \frac{13\sqrt{3}}{48} < \frac{1}{2}.$$

The last inequality follows from $3 \cdot 13^2 < 24^2$. A paired pair still has inner product $1/2$, as do its contacts with the appropriate poles. All other pole products are unchanged. $\square$

With the equator, poles, and labels fixed, the common direction of each pair is free to move in an open subset of S^{23} . The height is forced, but the direction is not. Thus saturation of the latitude constraints does not imply rigidity of the lifted configuration.

D.2. Independent motion of the dense layers. In dimensions $m = n = 38, 39$, the dense sign codes have minimum distance ten. Their mutual inner products are therefore at most $1 - 20/m < 1/2$. Their products with the middle and root layers are at most $\sqrt{8/m}$ and $\sqrt{2/m}$, also strictly less than $1/2$. Every dense point is an isolated vertex of the contact graph.

Proposition D.2. *For $m = n \in \{38, 39\}$, keep X_M and X_R fixed and put*

$$\eta_m = \frac{40 - m}{4m}.$$

Each $x \in X_T$ may independently be replaced by a unit vector $\hat{x}$ with $\|\hat{x} - x\| < \eta_m$. The resulting configuration has the same cardinality and satisfies the kissing condition. No new contact involves a dense point.

Proof. The change in a product of two unit vectors is at most the sum of their displacement lengths. Hence distinct moved dense points have inner product strictly less than

$$1 - \frac{20}{m} + 2\eta_m = \frac{1}{2}.$$

For a moved dense point and a fixed middle point, the bound is $\sqrt{8/m} + \eta_m < 1/2$. To verify the latter inequality exactly, the positive number $1/2 - \eta_m$ has square exceeding $8/m$ by $153/5776$ at $m = 38$ and by $937/24336$ at $m = 39$. The root-layer bound is smaller. The other pairs are unchanged. $\square$

The respective displacement radii are $1/76$ and $1/156$. The codes provide exact starting configurations, while their neighbourhoods need not retain a binary description. The two motion results exhibit distinct sources of flexibility: collective motion of a contacting pair in dimension 25, and independent motion of strictly separated points in dimensions 38 and 39.

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